



* BEGIN *

**65 nm Logic and Mixed-Mode Low-K Process
Interconnect Capacitance Model**

(Ver. 1.2_P. 2)

P-Sub, Logic and Mixed-Mode Process

DSM NO. : G-04-LOGIC/MIXED_MODE65N-LOW_K-INTERCAP

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2. Revision History

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
1.2	2	04/14/2016	1X Rs table Wd=0.03um	1X Rs table Wd=0.30um	

3. Interconnect Capacitance Model

3.1 Scope

There are 38 intercap model subsets classified by L65 BEOL metal stack option as follows:

G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1
G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V1.0_P1
G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
G-4L-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1
G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
G-4O-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_METAL3.6K-INTERCAP V0.1_P1

G-4A1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
G-4B1-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1
G-4E1-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
G-4G1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
G-4I1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
G-4J1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1
G-4K1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1
G-4L1-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1
G-4M1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
G-4N1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1
G-4O1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

G-4A2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
G-4B2-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1
G-4E2-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
G-4G2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
G-4I2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
G-4J2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1
G-4K2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1
G-4L2-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1
G-4M2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
G-4N2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1
G-4O2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

The according DSM naming (G01~G03) of these fifteen model subsets are listed as follows:

	Related DSM No.	INTERCAP
1P10M	G-01-LOGIC/MIXED_MODE65N-LL/LOW_K-DSM	G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_ME
	EDR : G-02-LOGIC/MIXED_MODE65N-LL/LOW_K-EDR	TAL8K-INTERCAP
	TLR : G-03-LOGIC/MIXED_MODE65N-LL-TLR	G-4A1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_M
	G-01-EDRAM65N-LL/LOW_K-DSM	ETAL8K/AL25K-INTERCAP
	EDR : G-02-EDRAM65N-EDR	G-4A2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_M
	TLR : G-03-EDRAM65N-LL/LOW_K-TLR	ETAL8K/AL36K-INTERCAP
	G-01-EFLASH/EE2PROM65N-LL/LOW_K-DSM	G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_ME
	EDR : G-02-EFLASH/EE2PROM65N-LL/LOW_K-EDR	TAL8K/CU_PAD-INTERCAP
	TLR : G-03-EFLASH/EE2PROM65N-LL-TLR	G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_
		METAL8K-INTERCAP
		G-4I1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP
		_METAL8K/AL25K-INTERCAP
		G-4I2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP
		_METAL8K/AL36K-INTERCAP
		G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_ME
		TAL32.5K-INTERCAP
		G-4J1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_M
		ETAL32.5K/AL25K-INTERCAP
		G-4J2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_M
		ETAL32.5K/AL36K-INTERCAP
		G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP
		_METAL32.5K-INTERCAP
		G-4K1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TO
		P_METAL32.5K/AL25K-INTERCAP
		G-4K2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TO
		P_METAL32.5K/AL36K-INTERCAP
		G-4L-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP
		_METAL12.5K-INTERCAP
		G-4L1-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TO
		P_METAL12.5K/AL25K-INTERCAP
		G-4L2-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TO
		P_METAL12.5K/AL36K-INTERCAP
	G-01-LOGIC/MIXED_MODE65N-SP/LOW_K-DSM	G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_ME
	EDR : G-02-LOGIC/MIXED_MODE65N-SP/LOW_K-EDR	TAL8K-INTERCAP
	TLR : G-03-LOGIC/MIXED_MODE65N-SP-TLR	G-4G1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_M
	G-01-EDRAM65N-SP/LOW_K-DSM	ETAL8K/AL25K-INTERCAP
	EDR : G-02-EDRAM65N-EDR	G-4G2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_M
	TLR : G-03-EDRAM65N-SP/LOW_K-TLR	ETAL8K/AL36K-INTERCAP
		G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_ME
		TAL8K/CU_PAD-INTERCAP
		G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TO
		P_METAL8K-INTERCAP
		G-4M1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/T
		OP_METAL8K/AL25K-INTERCAP
		G-4M2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/T
		OP_METAL8K/AL36K-INTERCAP
		G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_ME
		TAL32.5K-INTERCAP
		G-4N1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_M
		ETAL32.5K/AL25K-INTERCAP
		G-4N2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_M
		ETAL32.5K/AL36K-INTERCAP
		G-4O-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TO
		P_METAL32.5K-INTERCAP
		G-4O1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/T
		OP_METAL32.5K/AL25K-INTERCAP
		G-4O2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/T
		OP_METAL32.5K/AL36K-INTERCAP

	Related DSM No.	INTERCAP
1P9M2H	G-01-LOGIC/MIXED_MODE65N-SP/LOW_K-DSM	G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_M
	EDR : G-02-LOGIC/MIXED_MODE65N-SP/LOW_K-EDR	ETAL12.5K-INTERCAP
	TLR : G-03-LOGIC/MIXED_MODE65N-SP-TLR	G-4B1-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_M
	G-01-EDRAM65N-SP/LOW_K-DSM	METAL12.5K/AL25K-INTERCAP
	EDR : G-02-EDRAM65N-EDR	G-4B2-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_M
	TLR : G-03-EDRAM65N-SP/LOW_K-TLR	METAL12.5K/AL36K-INTERCAP
1P9M2F	G-01-LOGIC/MIXED_MODE65N-SP/LOW_K-DSM	G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_M
	EDR : G-02-LOGIC/MIXED_MODE65N-SP/LOW_K-EDR	ETAL12.5K/CU_PAD-INTERCAP
	TLR : G-03-LOGIC/MIXED_MODE65N-SP-TLR	G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_M
	G-01-EDRAM65N-SP/LOW_K-DSM	ETAL8K-INTERCAP
	EDR : G-02-EDRAM65N-EDR	G-4E1-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_M
	TLR : G-03-EDRAM65N-SP/LOW_K-TLR	METAL8K/AL25K-INTERCAP
		G-4E2-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_M
		METAL8K/AL36K-INTERCAP
		G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_M
		ETAL8K/CU_PAD-INTERCAP

	Related DSM No.	INTERCAP
1P8M1T	G-01-LOGIC/MIXED_MODE65N-LL/LOW_K-DSM	G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_M
	EDR : G-02-LOGIC/MIXED_MODE65N-LL/LOW_K-EDR	ETAL3.6K-INTERCAP
	TLR : G-03-LOGIC/MIXED_MODE65N-LL-TLR	
	G-01-EDRAM65N-LL/LOW_K-DSM	
	EDR : G-02-EDRAM65N-EDR	
	TLR : G-03-EDRAM65N-LL/LOW_K-TLR	
	G-01-EFLASH/EE2PROM65N-LL/LOW_K-DSM	
	EDR : G-02-EFLASH/EE2PROM65N-LL/LOW_K-EDR	
	TLR : G-03-EFLASH/EE2PROM65N-LL-TLR	

3.2 "Array Above A Ground Plane" Model

Interconnect capacitances are summarized to look-up tables for different kinds of conductor spacing. The sum capacitance (C_{sum}) can be derived from C_a , C_f and C_c .

$$C_{sum} = C_a + 2 \cdot C_f + 2 \cdot C_c \quad (\text{unit: fF/um})$$

where

W : Conductor width (unit: um).

S : Conductor spacing (unit: um).

C_a : Parallel plate area capacitance (unit: fF/um).

C_f : Fringing capacitance to bottom plane (unit: fF/um/side).

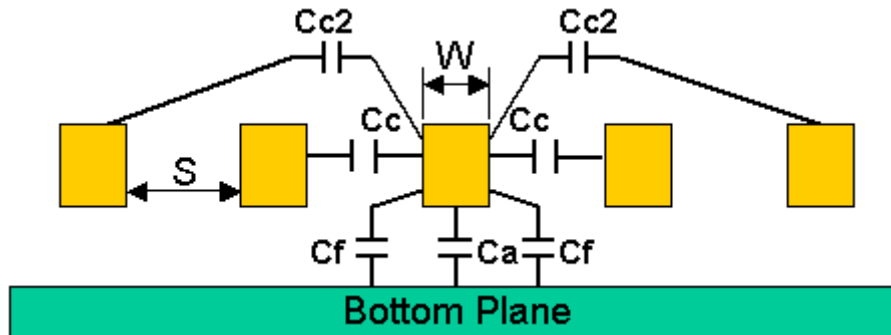
C_c : Coupling capacitance to adjacent conductor (unit: fF/um/side).

$$C_{total} = C_{sum} + 2 \cdot C_{c2} \quad (\text{unit: fF/um})$$

where

C_{c2} : Coupling capacitance to outer conductor (unit: fF/um/side).

The difference between C_{sum} and C_{total} is the effect of the coupling capacitance from the two blocks that far from the center.



3.3 "Array Between Ground Planes" Model

The sum capacitance (C_{sum}) can be derived from C_{atop} , C_{abot} , C_{su} , C_{sd} and C_c .

$$C_{sum} = (C_{atop} + C_{abot}) + 2*(C_{su} + C_{sd}) + 2*C_c \text{ (unit: fF/um)}$$

where

W : Conductor width (unit: um).

S : Conductor spacing (unit: um).

C_{atop} : Capacitance of center conductor to top ground plane (unit: fF/um).

C_{abot} : Capacitance of center conductor to bottom ground plane (unit: fF/um).

C_{su} : Side-wall up capacitance coefficient (unit: fF/um/side).

C_{sd} : Side-wall down capacitance coefficient (unit: fF/um/side).

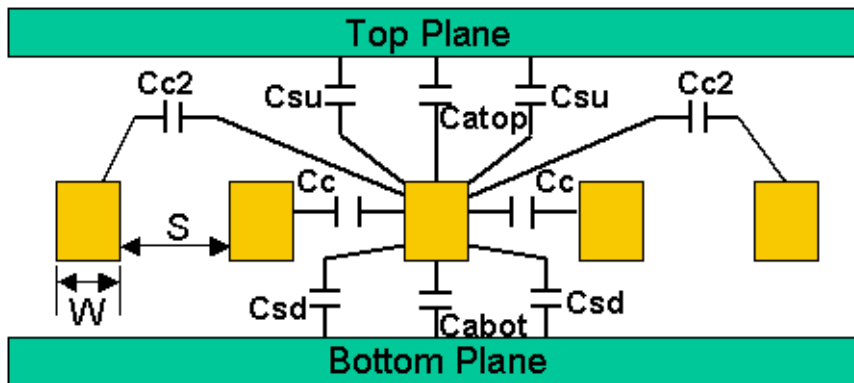
C_c : Coupling capacitance to adjacent conductor (unit: fF/um/side)

$$C_{total} = C_{sum} + 2*C_{c2} \text{ (unit: fF/um)}$$

where

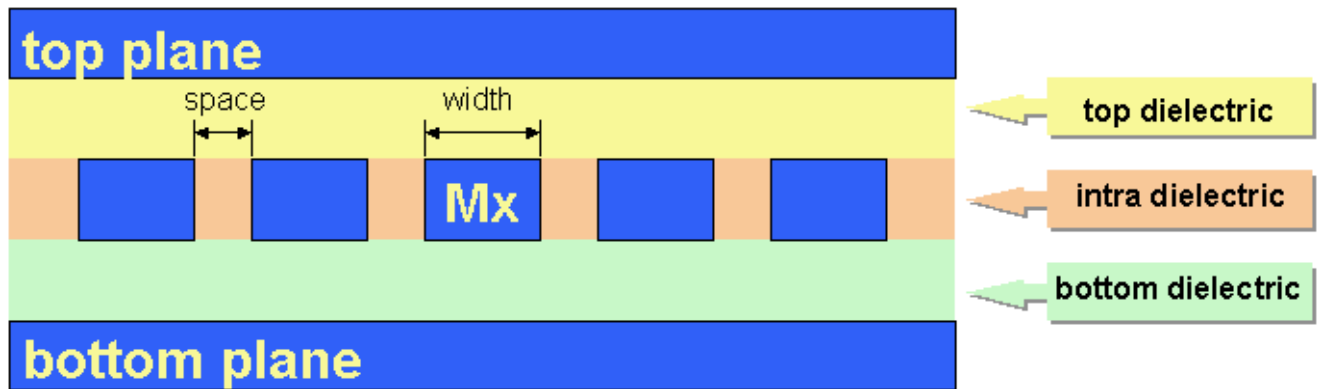
C_{c2} : Coupling capacitance to outer conductor (unit : fF/um/side).

The difference between C_{sum} and C_{total} is the effect of the coupling capacitance from the two blocks that far from the center.



3.4 Corner Model Methodology

The corner models (typical case, minimum case and maximum case) are based on a fixed pitch and the variations of metal width and top/intra/bottom dielectric thickness. Metal width is separated to typical metal width, minimum metal width and maximum metal width. The top/intra/bottom dielectric thicknesses are separated to typical top/intra/bottom dielectric thickness, minimum top/intra/bottom dielectric thicknesses and maximum top/intra/bottom dielectric thicknesses. So, the metal width and dielectric thickness for each corner model are listed as follows:



A. Typical Case :

1. Metal Width : typical metal width
2. Dielectric Thickness : typical thicknesses of top/intra/bottom dielectric

B. Minimum Case :

1. Metal Width : minimum metal width
2. Dielectric Thickness : minimum thicknesses of intra dielectric, maximum thicknesses of top/bottom dielectric

C. Maximum Case :

1. Metal Width : maximum metal width
2. Dielectric Thickness : maximum thicknesses of intra dielectric, minimum thicknesses of top/bottom dielectric

Where typical metal width = physical metal width

minimum metal width = physical metal width – metal width variation

maximum metal width = physical metal width + metal width variation

typical dielectric thickness = physical dielectric thickness

minimum dielectric thickness = physical dielectric thickness – dielectric thickness variation

maximum dielectric thickness = physical dielectric thickness + dielectric thickness variation

4. Process Parameters

4.1 Simulated Thickness and Width of Each Conductor Layer

G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4A1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4A2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4B1-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4B2-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V1.0_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4E1-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4E2-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

- Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010

- Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4G1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4G2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4I1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4I2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
 2. The ΔW value is based on WEE table of each conductor.

G-4J1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4J2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4K1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4K2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4L-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal10	1.250	1.063	1.438	drawing width+ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4L1-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal10	1.250	1.063	1.438	drawing width+ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4L2-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	1.250	1.063	1.438	drawing width +ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
metal10	1.250	1.063	1.438	drawing width+ΔW	± 0.056	0.560	0.560	0.560	0.560	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4M1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4M2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4N1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4N2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-40-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
2. The ΔW value is based on WEE table of each conductor.

G-401-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	2.500	2.125	2.875	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.
2. The ΔW value is based on WEE table of each conductor.

G-4O2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.02	± 0.004	0.060	0.120	0.040	0.140	-0.020
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
MMCBP	0.110	-	-	drawing width	± 0.15	-	-	-	-	-
MMCTP	0.060	-	-	drawing width	± 0.15	-	-	-	-	-
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
metal9	0.800	0.680	0.920	drawing width +ΔW	± 0.041	0.400	0.400	0.410	0.390	0.010
metal10	3.250	2.763	3.738	drawing width	± 0.2	2.000	2.000	2.000	2.000	0.000
AL-RDL	3.600	3.060	4.140	drawing width	± 0.27	2.700	1.800	2.700	1.800	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_METAL3.6K-INTERCAP V0.1_P1

Conductor	Thickness (um)			Width (um)		Drawing and physical dimension in ground rule (um)				
	Typ	Min	Max	Physical width	Variation	Wd_min	Sd_min	Wp_min	Sp_min	Width bias
poly1	0.100	0.085	0.115	drawing width -0.003	± 0.0057	0.060	0.120	0.057	0.123	-0.003
metal1	0.180	0.153	0.207	drawing width +ΔW	± 0.0095	0.090	0.090	0.095	0.085	0.005
metal2	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal3	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal4	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal5	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal6	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal7	0.220	0.187	0.253	drawing width +ΔW	± 0.01	0.100	0.100	0.100	0.100	0.000
metal8	0.360	0.306	0.414	drawing width +ΔW	± 0.0203	0.200	0.200	0.203	0.197	0.003
AL-RDL	1.200	1.020	1.380	drawing width	± 0.27	2.700	1.600	2.700	1.600	0.000

Note: 1. Metal thickness variation is +/- 15% for each metal layer in intercap model simulation.

2. The ΔW value is based on WEE table of each conductor.

4.2 Dielectric Thickness Variation

Dielectric thickness variation is +/- 15% for each dielectric layer in intercap model simulation.

4.3 WEE Table

The width bias (ΔW) of each conductor width is listed as following Wire Edge Enlargement tables. The tables include drawing width (Wd) and drawing space (Sd) for 1X rule ~ 4X rule of each metal layer.

(1) Metal1:

Sd \ Wd	0.09(um)	0.18(um)	0.27(um)	0.36(um)
0.09(um)	0.005	-0.018	-0.022	-0.026
0.18(um)	0.026	0.001	0.001	-0.005
0.27(um)	0.039	-0.008	-0.009	-0.020
0.36(um)	0.051	-0.014	-0.014	-0.026

(2) Metal2 ~ Metal6 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2, G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4L, G4L1, G4L2, G4M, G4M1, G4M2, G4N G4N1, G4N2 and G4O, G4O1, G4O2:

Metal2 ~ Metal7 of G4B, G4B1, G4B2 and G4D, G4E, G4E1, G4E2, G4F, G4P:

Sd \ Wd	0.10(um)	0.20(um)	0.30(um)	0.40(um)
0.10(um)	0.000	-0.018	-0.016	-0.018
0.20(um)	0.026	0.008	0.018	0.016
0.30(um)	0.040	0.013	0.019	0.025
0.40(um)	0.049	0.017	0.019	0.028

(3) Metal7 and Metal8 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2, G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4L, G4L1, G4L2, G4M, G4M1, G4M2, G4N G4N1, G4N2 and G4O, G4O1, G4O2:

Metal8 of G4P:

Sd \ Wd	0.20(um)	0.40(um)	0.60(um)	0.80(um)
0.20(um)	0.003	-0.030	-0.032	-0.029
0.40(um)	0.025	-0.002	0.009	0.011
0.60(um)	0.020	-0.003	0.011	0.011
0.80(um)	0.020	-0.007	0.010	0.017

(4) Metal9 and Metal10 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2 and G4M, G4M1, G4M2:

Metal8 and Metal9 of G4E, G4E1, G4E2 and G4F:

Metal9 of G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4N, G4N1, G4N2 and G4O, G4O1, G4O2:

Sd \ Wd	0.40(um)	0.80(um)	1.20(um)	1.60(um)
0.40(um)	0.010	-0.030	-0.040	-0.050
0.80(um)	0.000	-0.040	-0.050	-0.060
1.20(um)	0.000	-0.040	-0.050	-0.060
1.60(um)	-0.010	-0.050	-0.050	-0.070

(5) Metal8 and Metal9 of G4B, G4B1, G4B2 and G4D:

Metal9 and Metal10 of G4L, G4L1, G4L2:

Sd \ Wd	0.56(um)	1.12(um)	1.68(um)	2.24(um)
0.56(um)	0.000	-0.024	-0.052	-0.068
1.12(um)	0.007	-0.009	-0.024	-0.051
1.68(um)	-0.001	-0.005	-0.031	-0.078
2.24(um)	-0.004	-0.003	-0.019	-0.052

4.4 Rs Table

The sheet resistance (Rs) of each conductor metal listed below is in units of $\Omega/\square$. Here, $\square$ is calculated by the drawn length divided drawn width.

(1) Metal1:

Sd \ Wd	0.09(um)	0.18(um)	0.27(um)	0.36(um)
0.09(um)	0.170	0.169	0.154	0.144
0.18(um)	0.129	0.148	0.140	0.134
0.27(um)	0.108	0.150	0.141	0.137
0.36(um)	0.096	0.154	0.142	0.139

(2) Metal2 ~ Metal6 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2, G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4L, G4L1, G4L2, G4M, G4M1, G4M2, G4N G4N1, G4N2 and G4O, G4O1, G4O2:
Metal2 ~ Metal7 of G4B, G4B1, G4B2 and G4D, G4E, G4E1, G4E2, G4F, G4P:

Sd \ Wd	0.10(um)	0.20(um)	0.30(um)	0.40(um)
0.10(um)	0.150	0.136	0.116	0.108
0.20(um)	0.114	0.118	0.105	0.101
0.30(um)	0.098	0.116	0.105	0.099
0.40(um)	0.091	0.115	0.105	0.099

(3) Metal7 and Metal8 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2, G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4L, G4L1, G4L2, G4M, G4M1, G4M2, G4N G4N1, G4N2 and G4O, G4O1, G4O2:

Metal8 of G4P:

Sd \ Wd	0.20(um)	0.40(um)	0.60(um)	0.80(um)
0.20(um)	0.070	0.073	0.070	0.069
0.40(um)	0.062	0.068	0.066	0.067
0.60(um)	0.063	0.068	0.067	0.067
0.80(um)	0.063	0.069	0.067	0.067

(4) Metal9 and Metal10 of G4A, G4A1, G4A2, G4C, G4G, G4G1, G4G2, G4H, G4I, G4I1, G4I2 and G4M, G4M1, G4M2:

Metal8 and Metal9 of G4E, G4E1, G4E2 and G4F:

Metal9 of G4J, G4J1, G4J2, G4K, G4K1, G4K2, G4N, G4N1, G4N2 and G4O, G4O1, G4O2:

Sd \ Wd	0.40(um)	0.80(um)	1.20(um)	1.60(um)
0.40(um)	0.027	0.028	0.027	0.027
0.80(um)	0.027	0.028	0.028	0.027
1.20(um)	0.028	0.028	0.028	0.028
1.60(um)	0.028	0.029	0.028	0.028

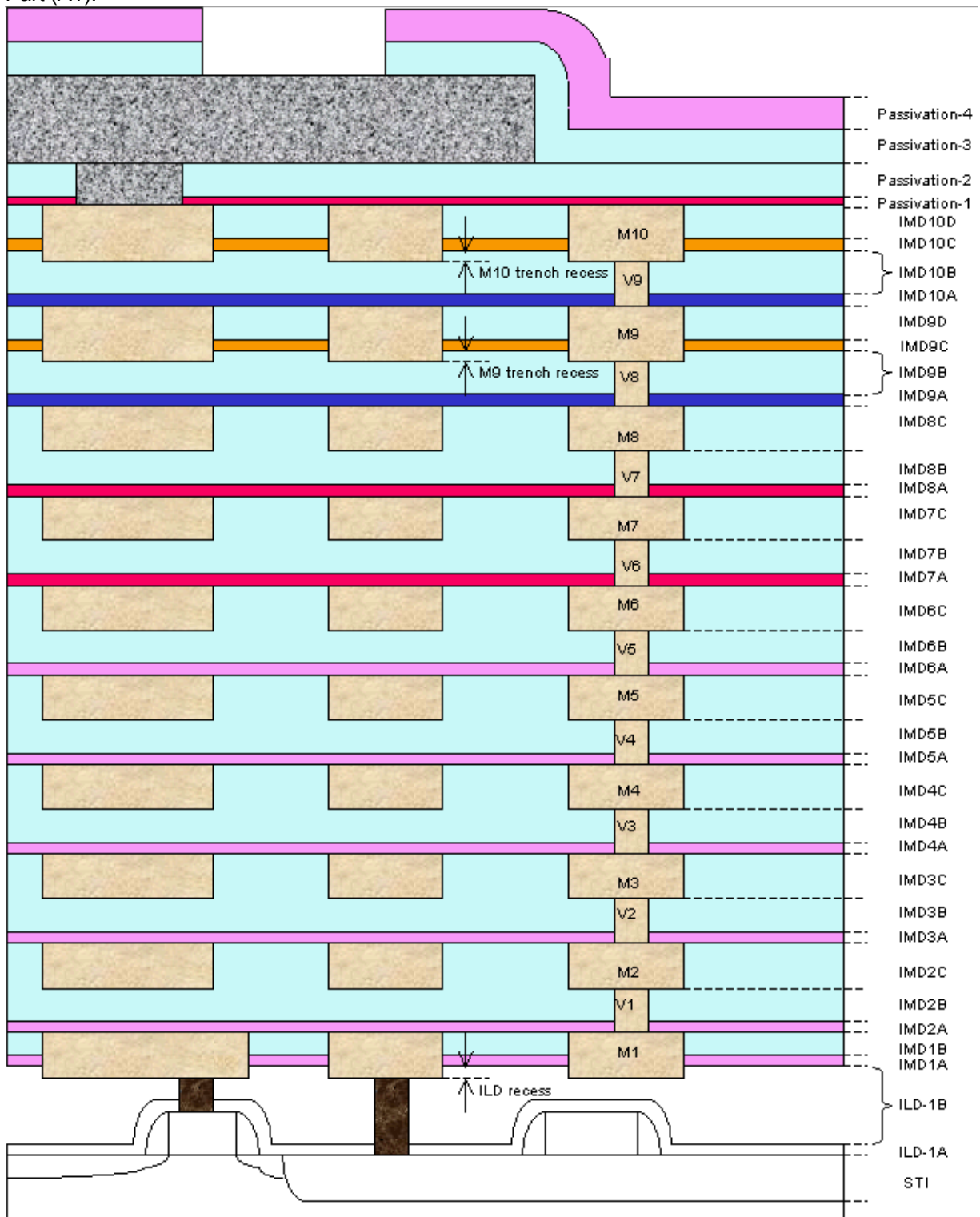
(5) Metal8 and Metal9 of G4B, G4B1, G4B2 and G4D:

Metal9 and Metal10 of G4L, G4L1, G4L2:

Sd \ Wd	0.56(um)	1.12(um)	1.68(um)	2.24(um)
0.56(um)	0.017	0.016	0.016	0.017
1.12(um)	0.017	0.016	0.017	0.017
1.68(um)	0.017	0.017	0.016	0.016
2.24(um)	0.017	0.017	0.016	0.017

4.5 Cross Section

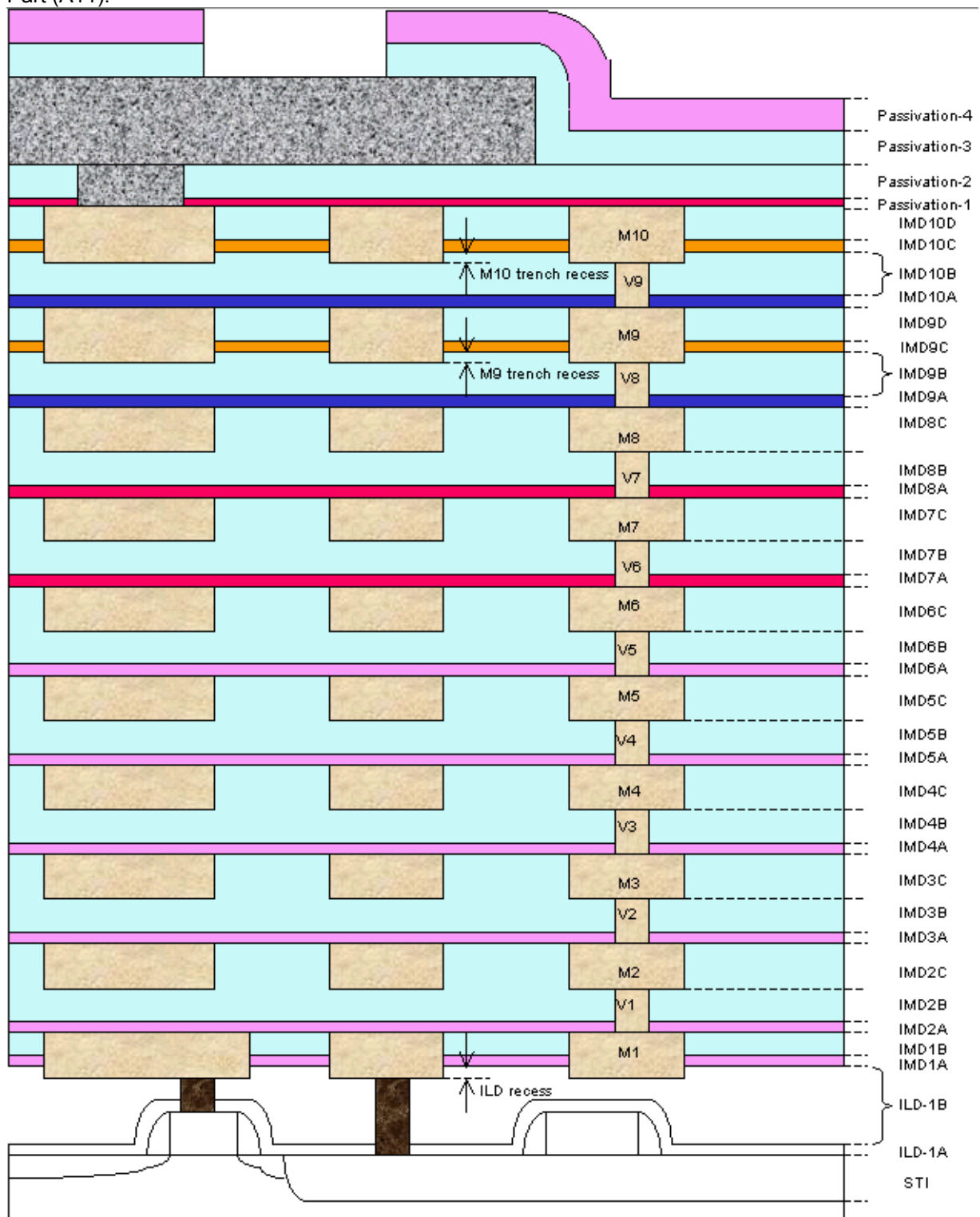
G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
Part (A1):



Part (A2):

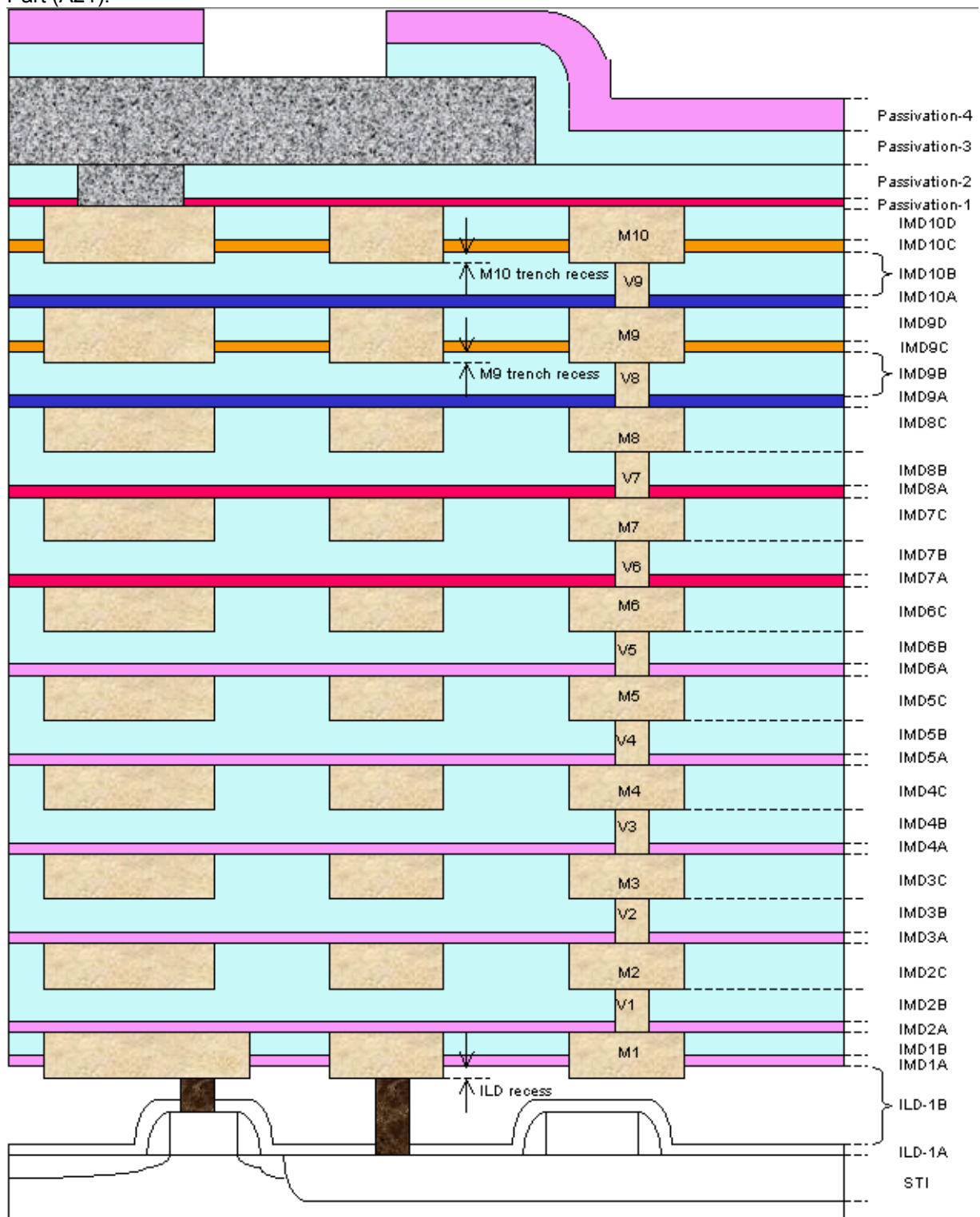
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4A1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
Part (A11):



Part (A12):

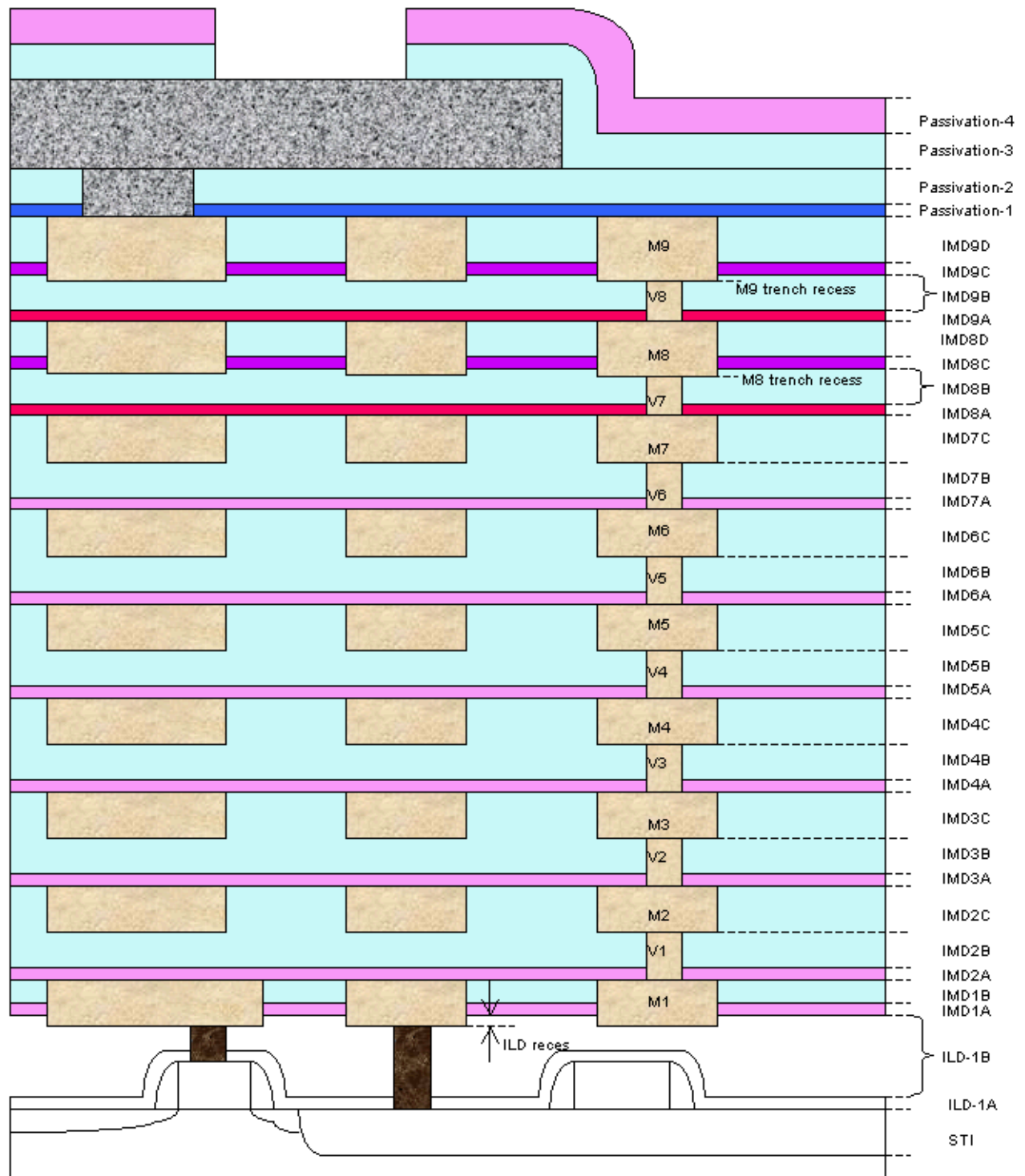
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4A2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
Part (A21):

Part (A22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

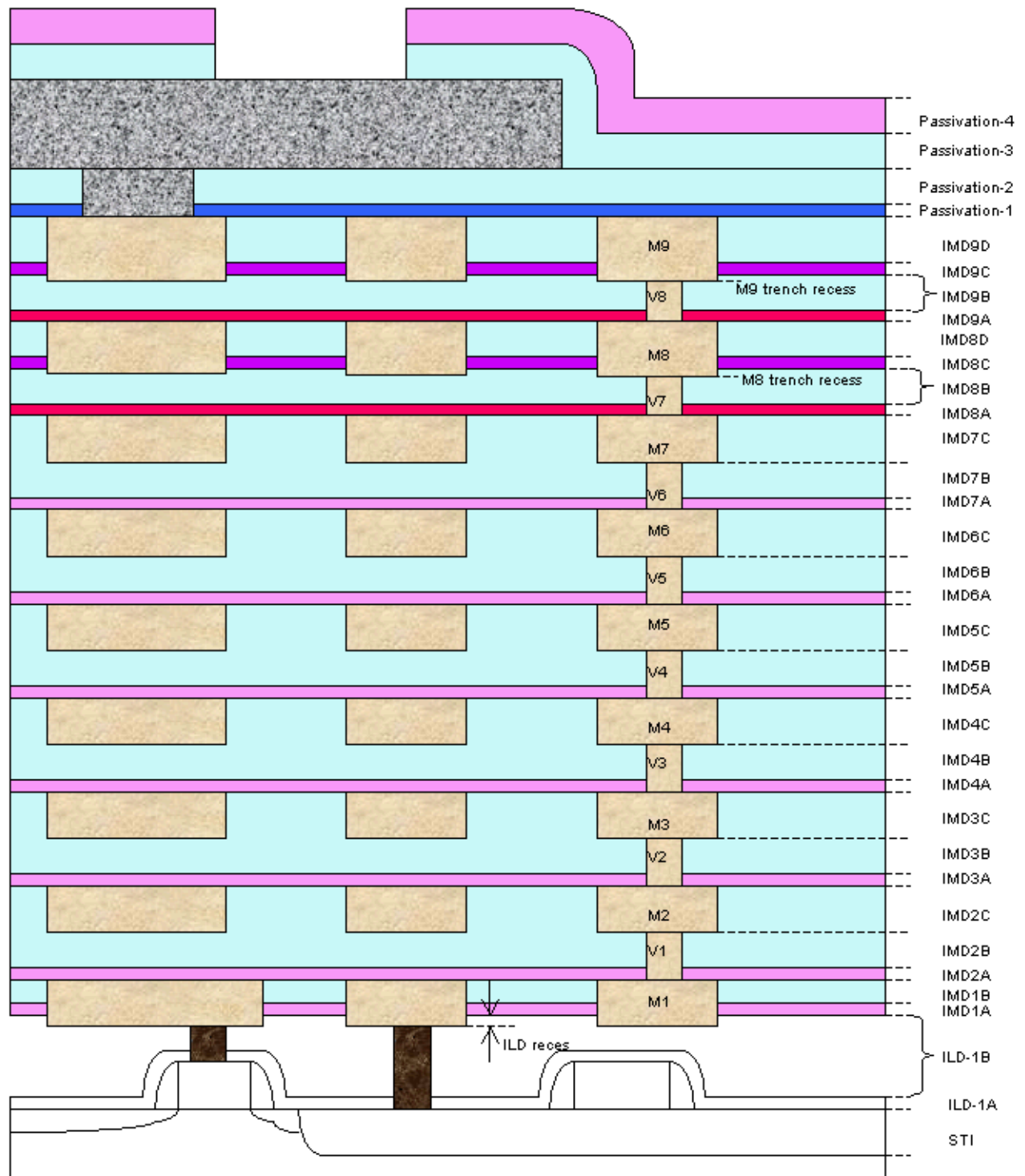
G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1
Part (B1):



Part (B2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.235	7
IMD8B	0.4	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	1.14	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4B1-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP
 V0.1_P1
 Part (B11):



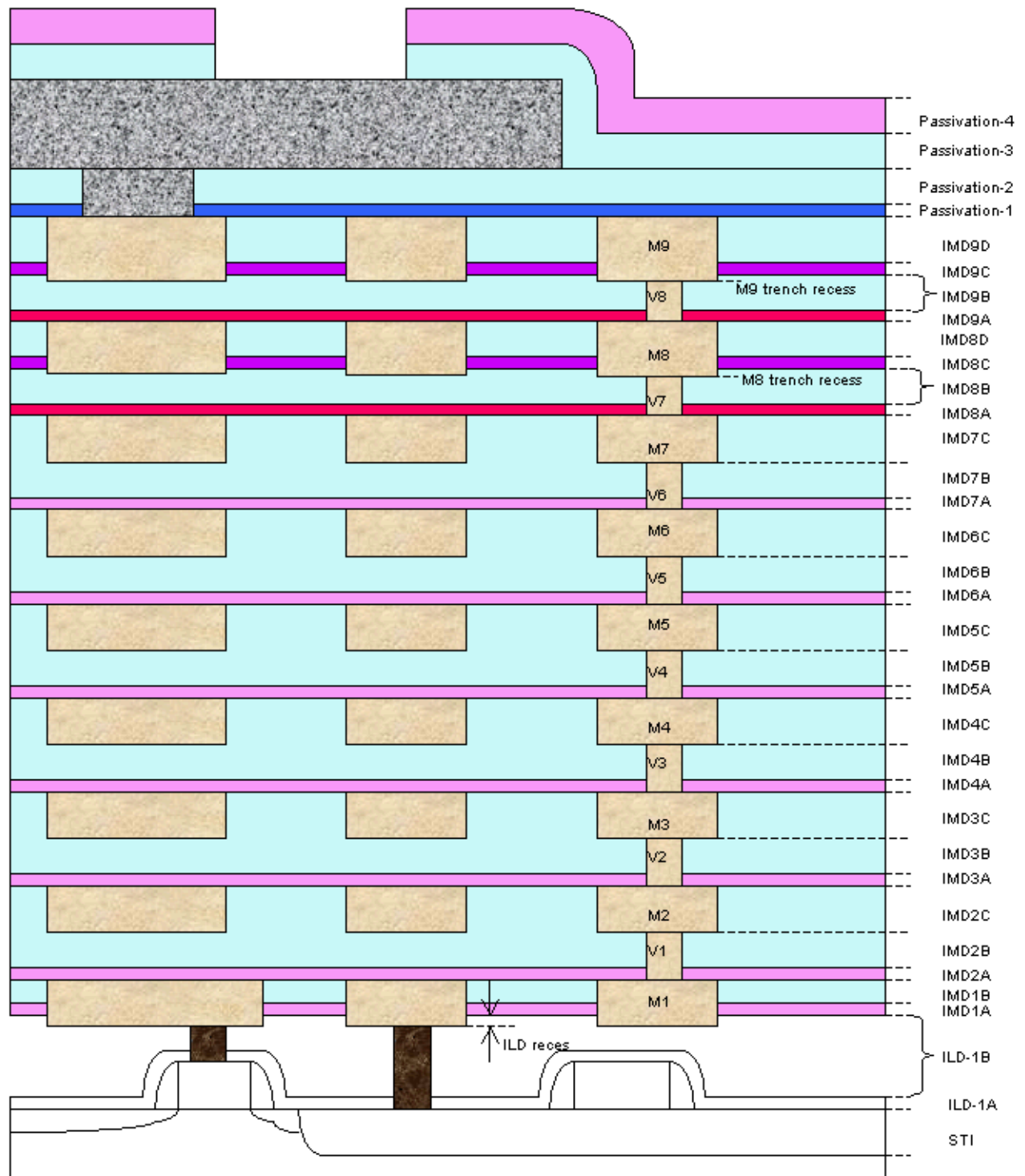
Part (B12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.235	7
IMD8B	0.4	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	1.14	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4B2-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP

V0.1_P1

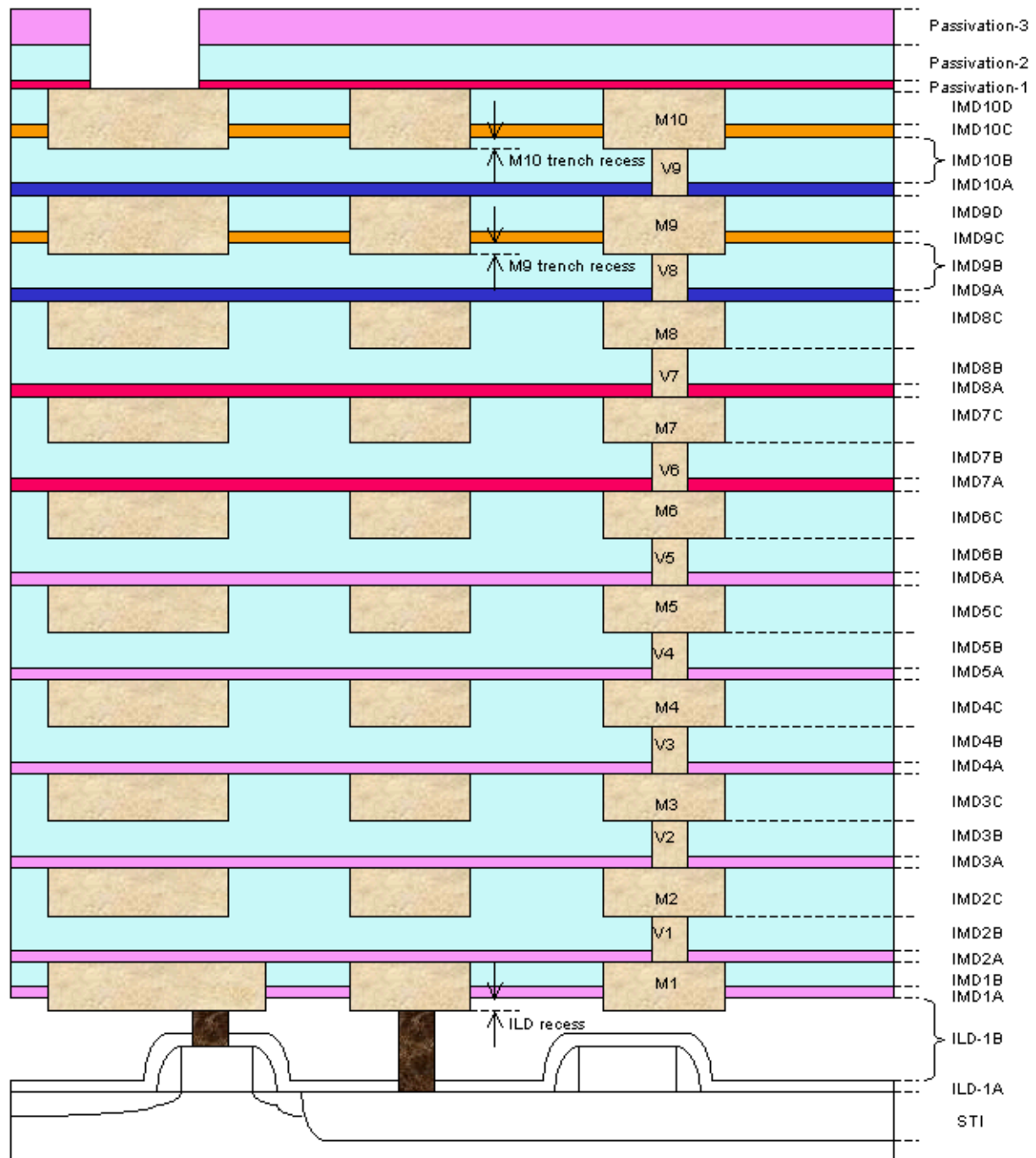
Part (B21):



Part (B22):

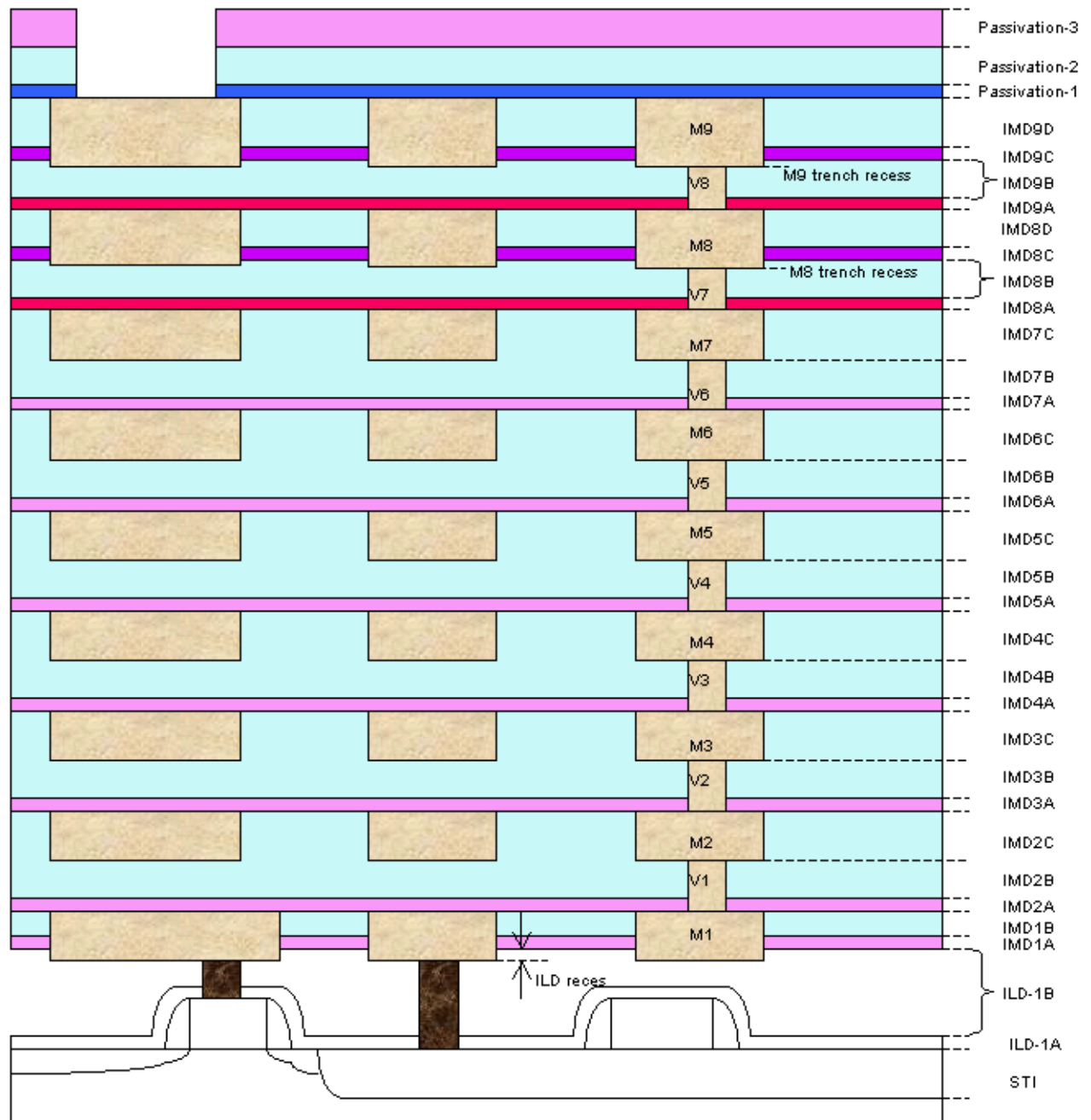
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.235	7
IMD8B	0.4	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	1.14	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
Part (C1):



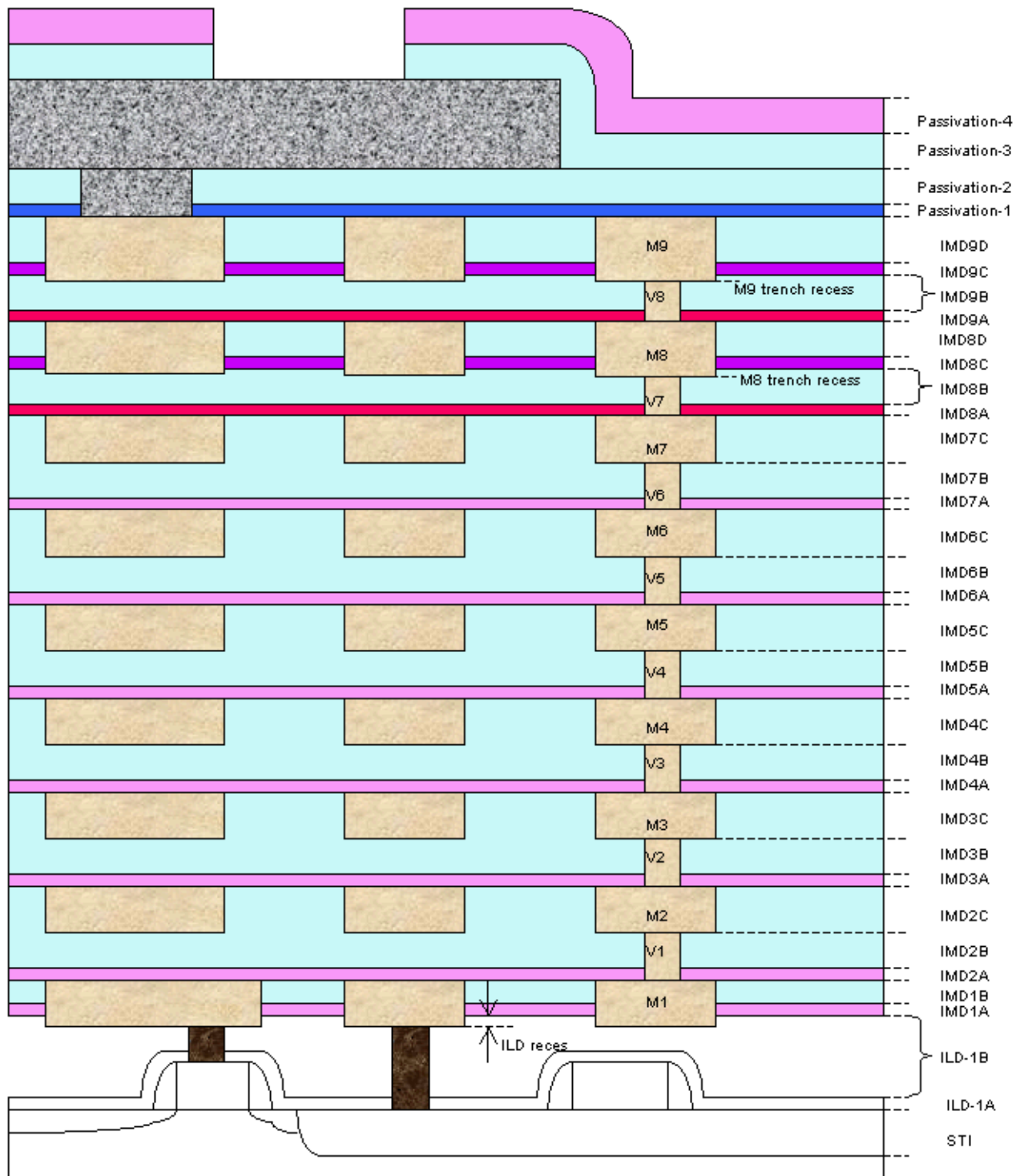
Part (C2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	0.4	4.1
Passivation3	0.5	7

G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V1.0_P1
Part (D1):

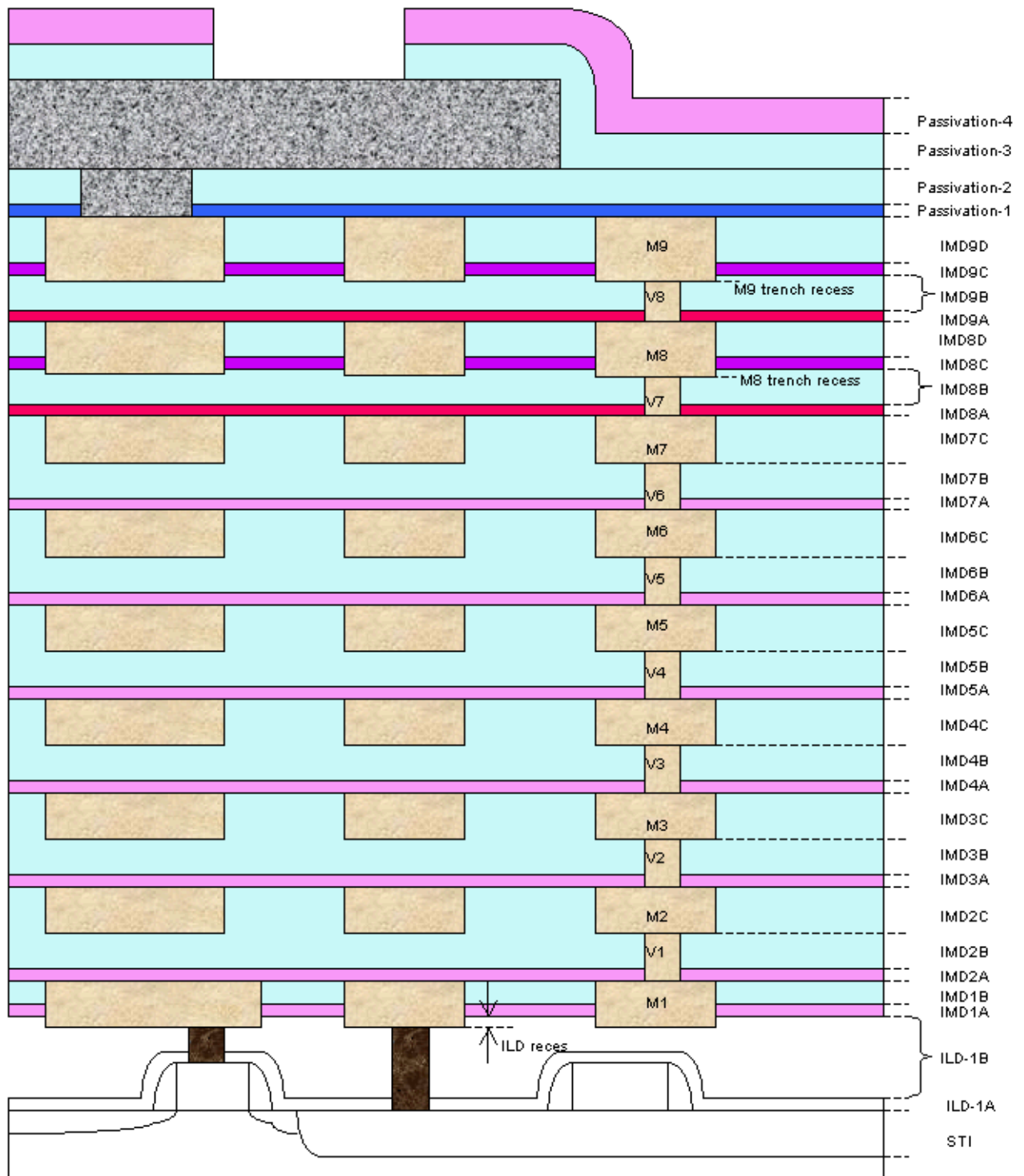
Part (D2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.235	7
IMD8B	0.4	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	1.14	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
Passivation1	0.07	7
Passivation2	0.4	4.1
Passivation3	0.5	7

G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
Part (E1):

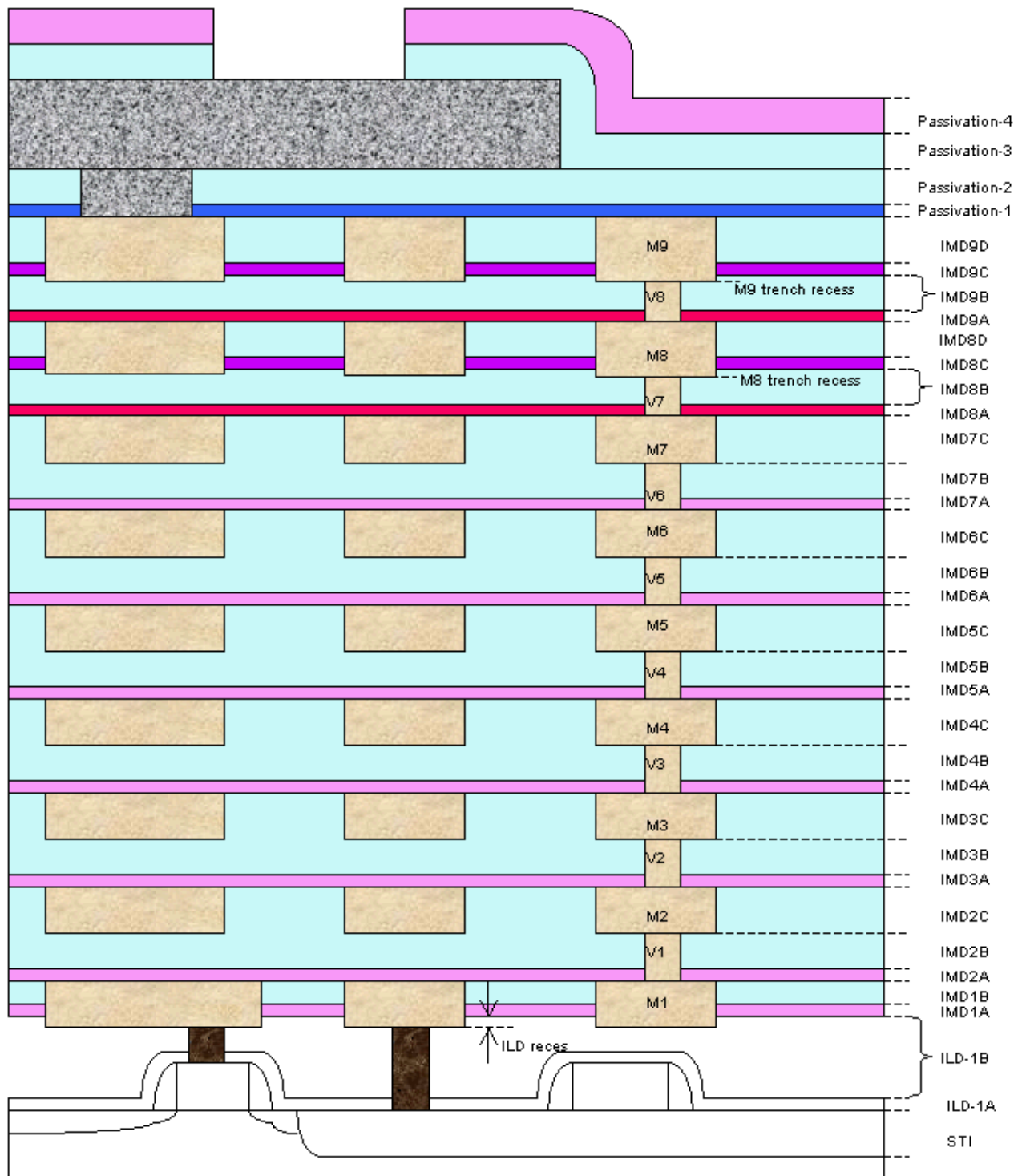
Part (E2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.135	7
IMD8B	0.5	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	0.69	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4E1-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
Part (E11):

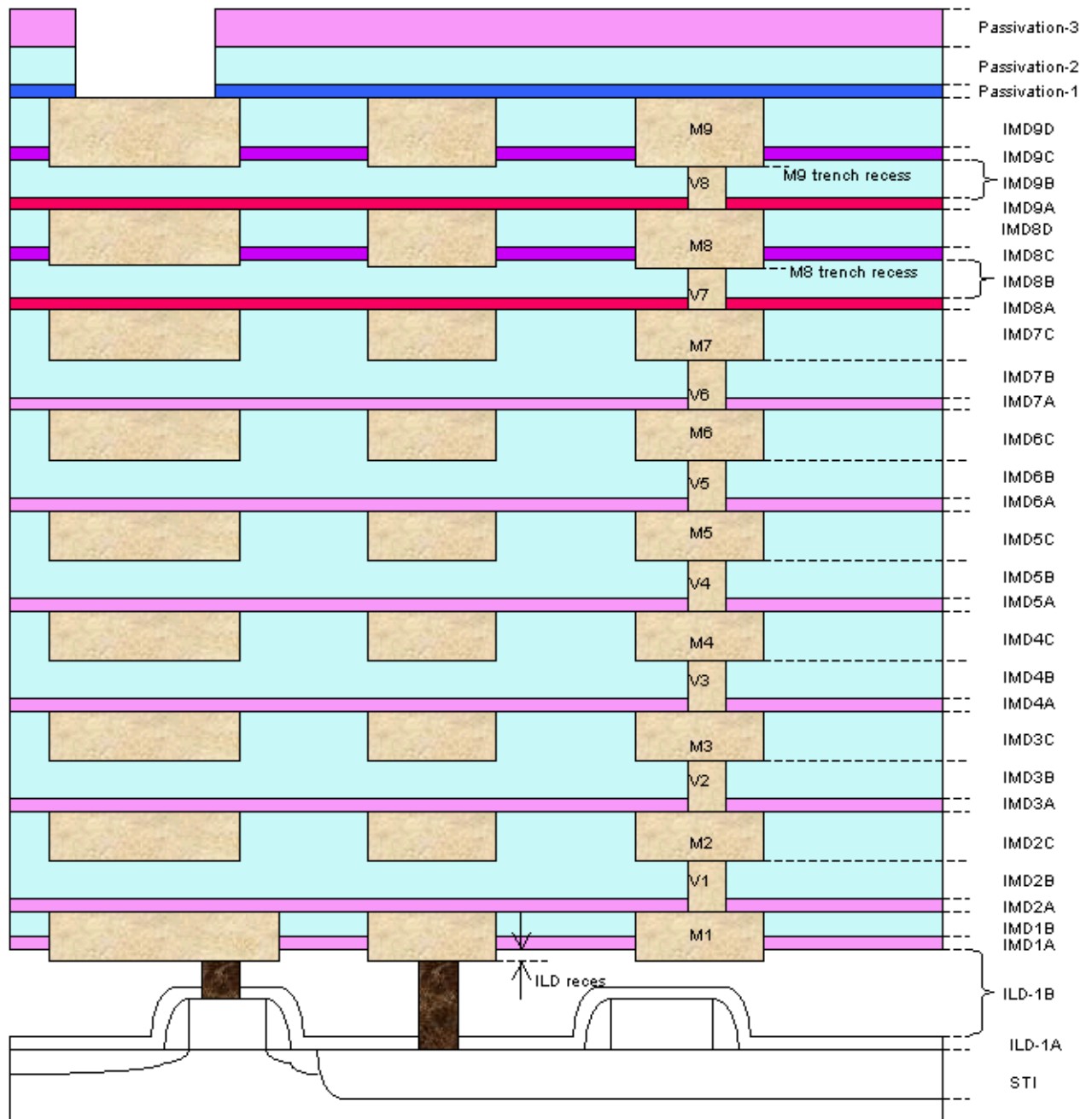
Part (E12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.135	7
IMD8B	0.5	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	0.69	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4E2-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
Part (E21):

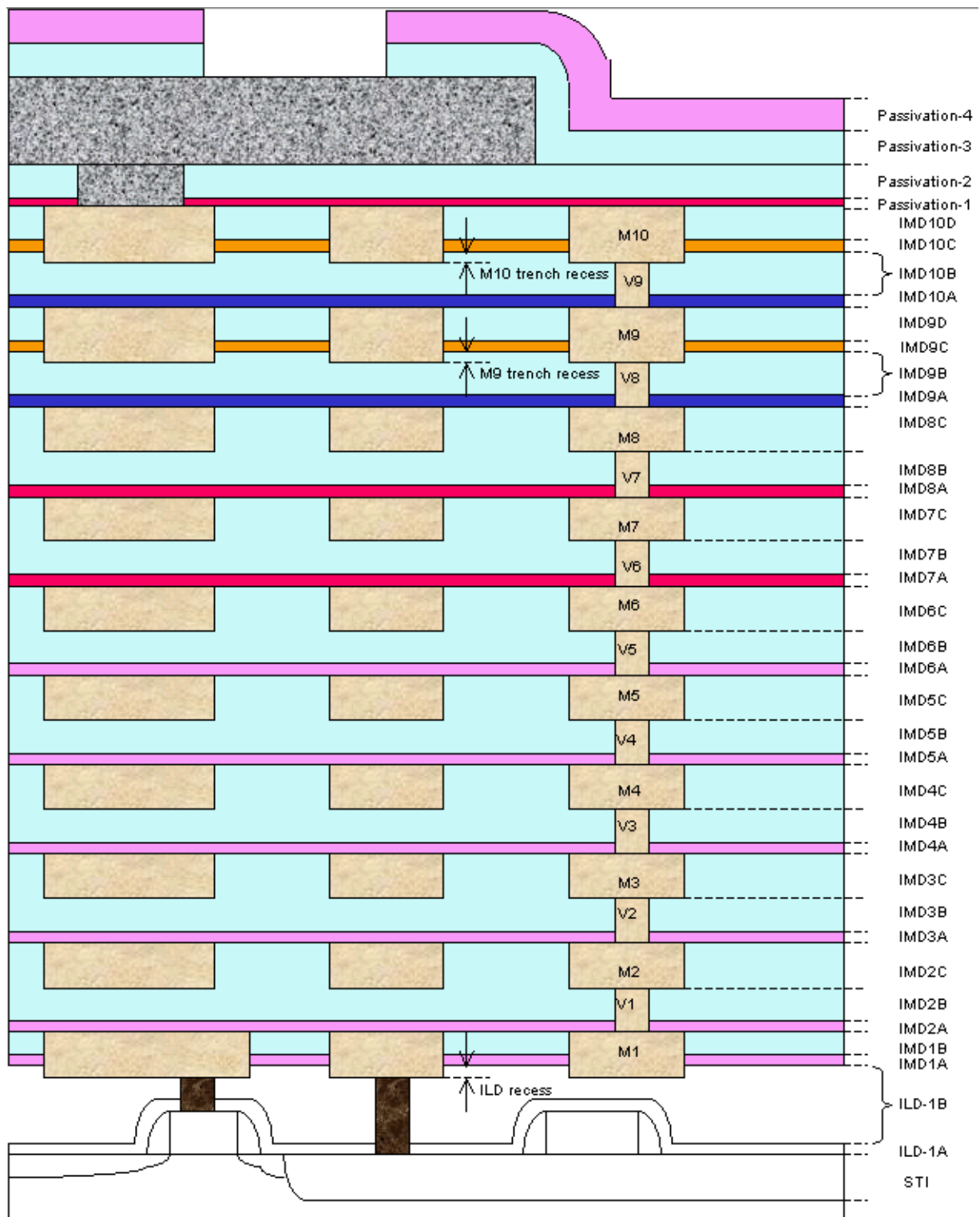
Part (E22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.135	7
IMD8B	0.5	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	0.69	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
Part (F1):

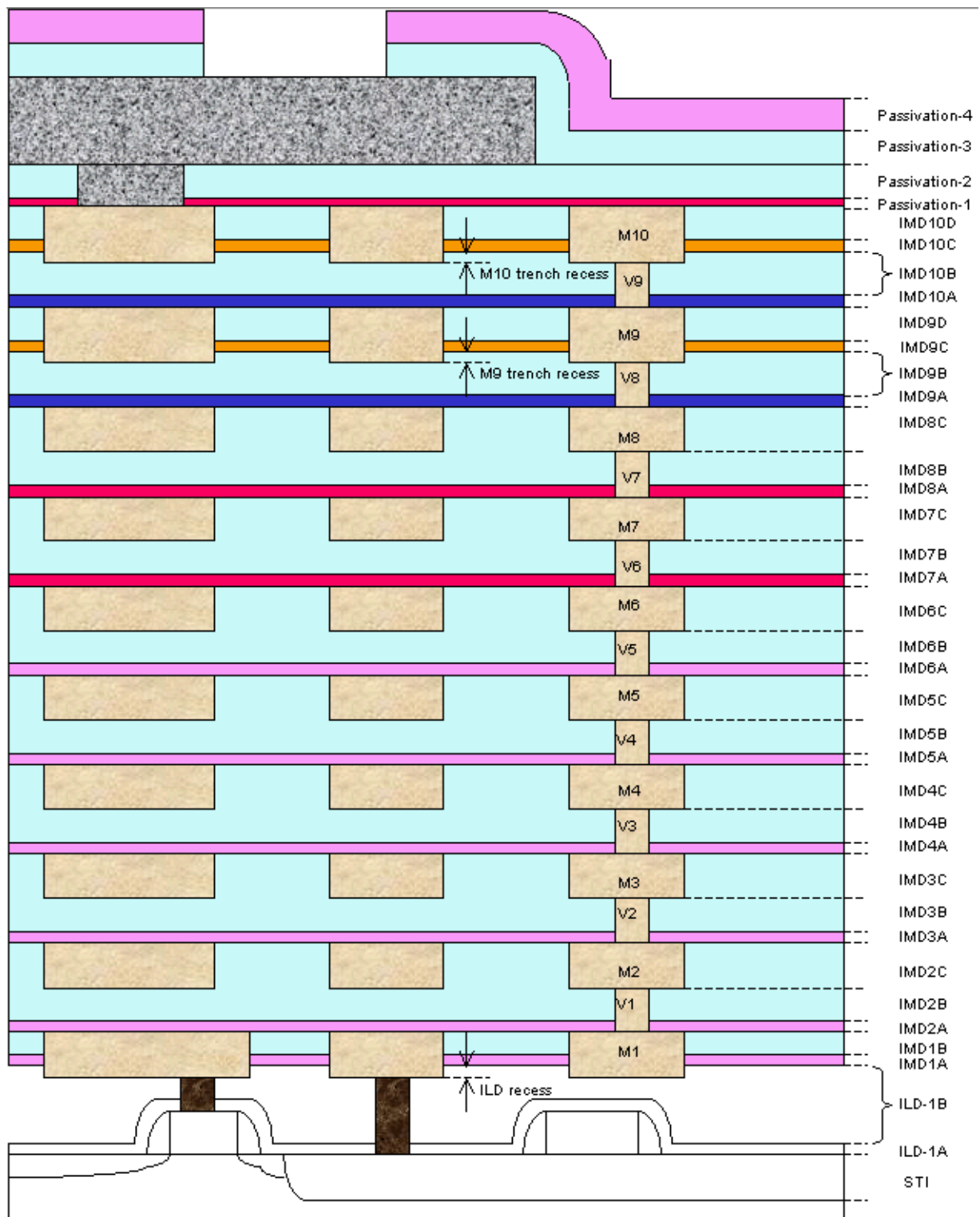
Part (F2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.135	7
IMD8B	0.5	3.7
M8 trench recess	0.02	3.7
IMD8C	0.09	5
IMD8D	0.69	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
Passivation1	0.07	7
Passivation2	0.4	4.1
Passivation3	0.5	7

G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
Part (G1):

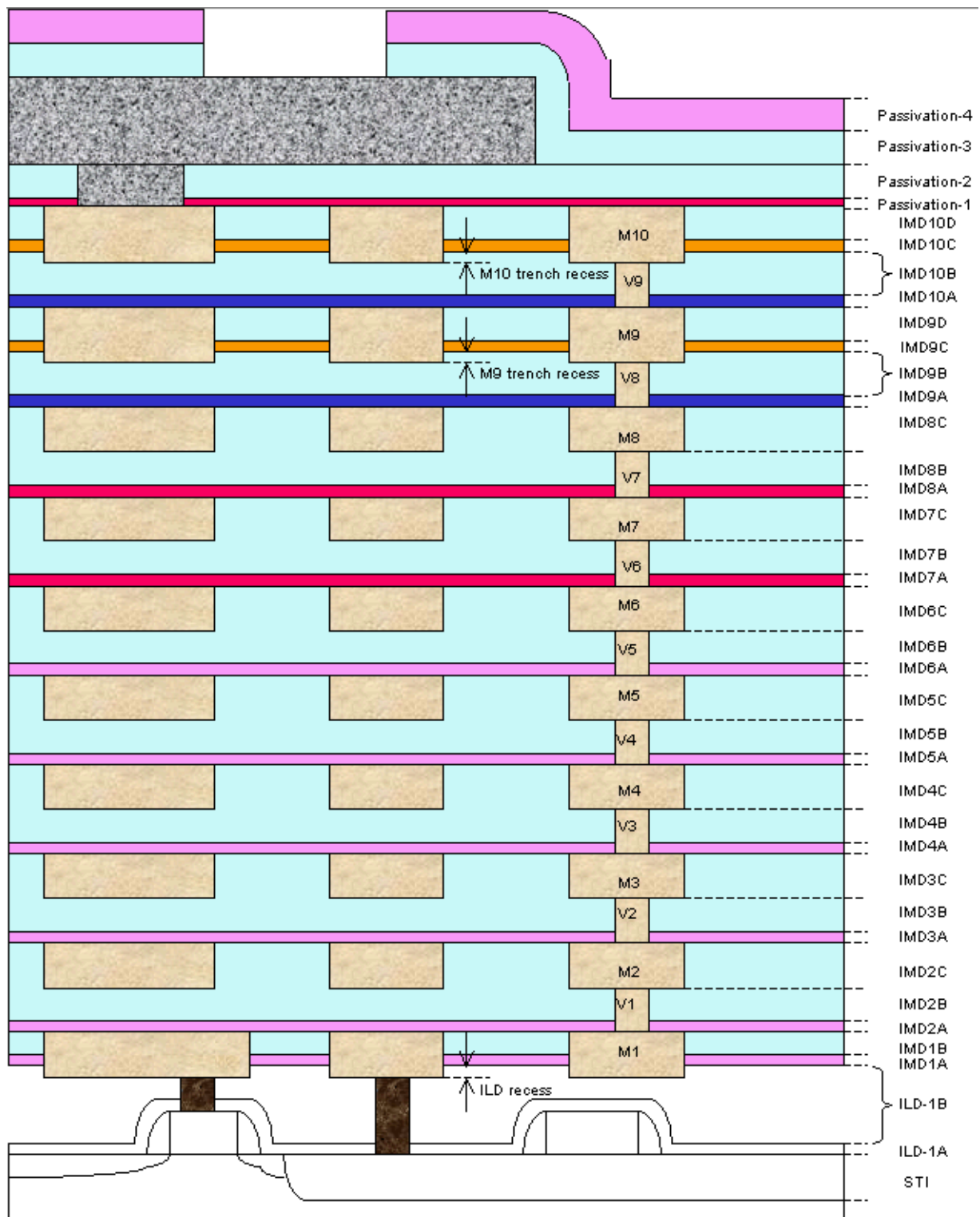
Part (G2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4G1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1
Part (G11):

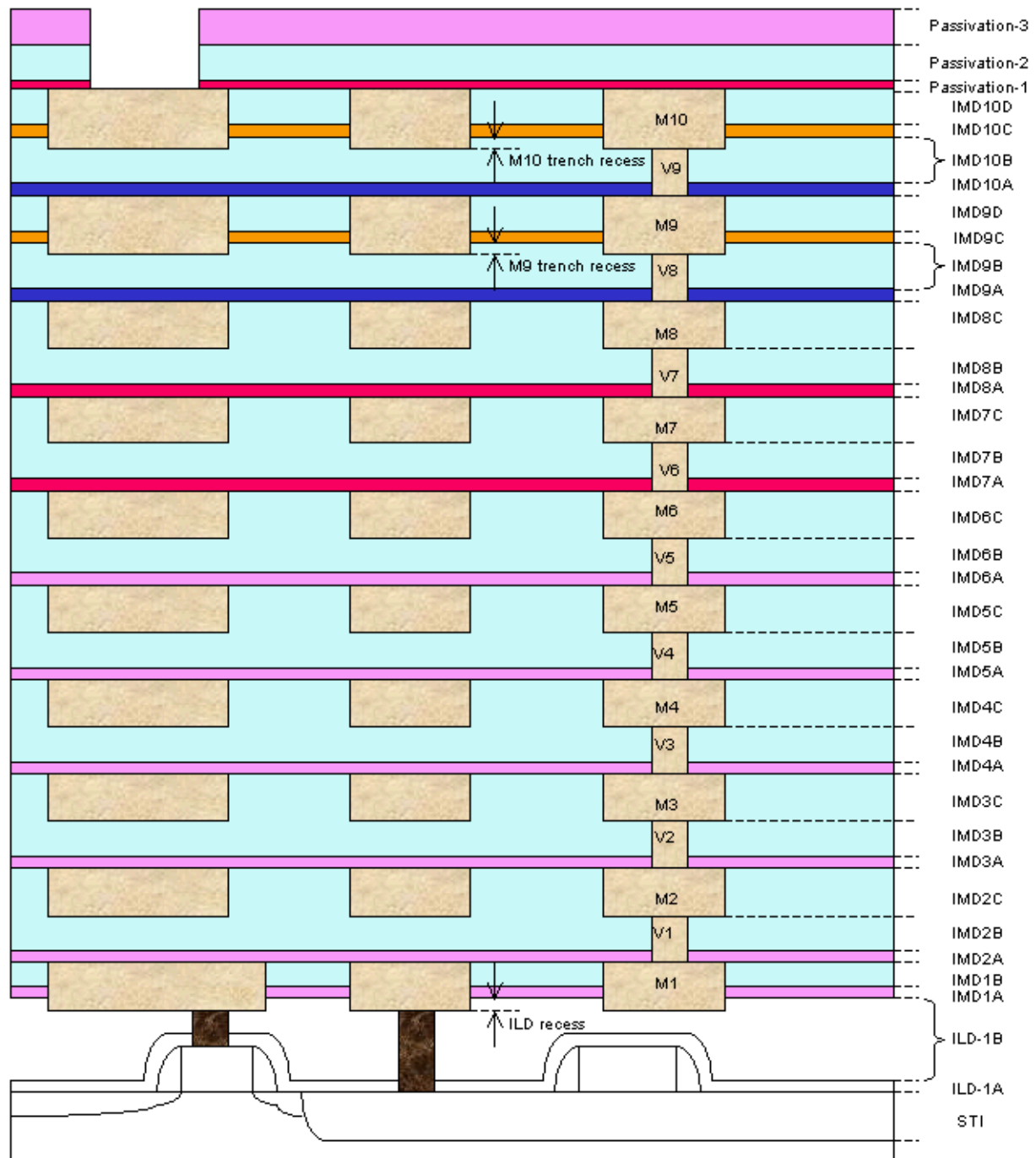
Part (G12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4G2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1
Part (G21):

Part (G22):

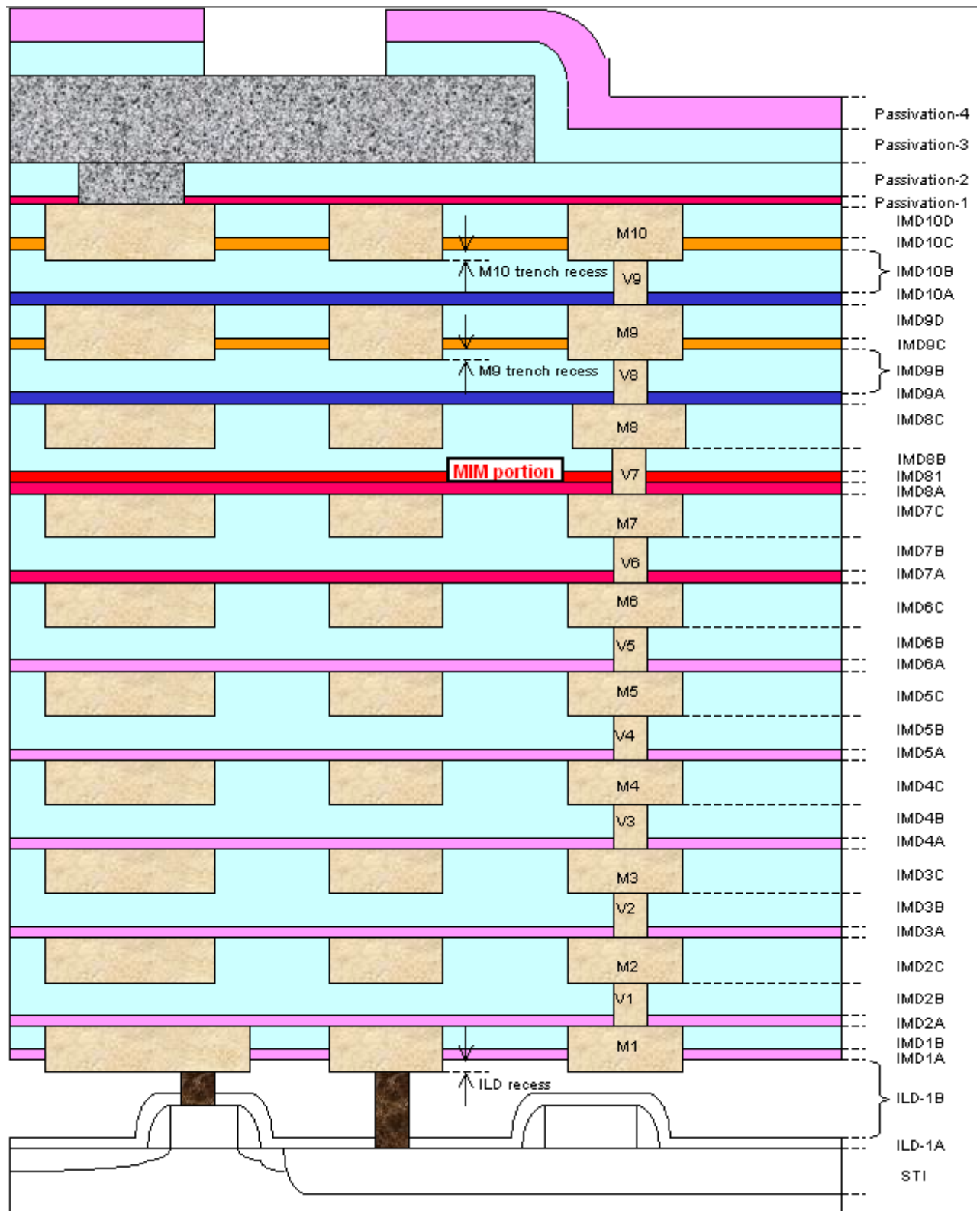
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1
Part (H1):

Part (H2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	0.4	4.1
Passivation3	0.5	7

G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
Part (I1):



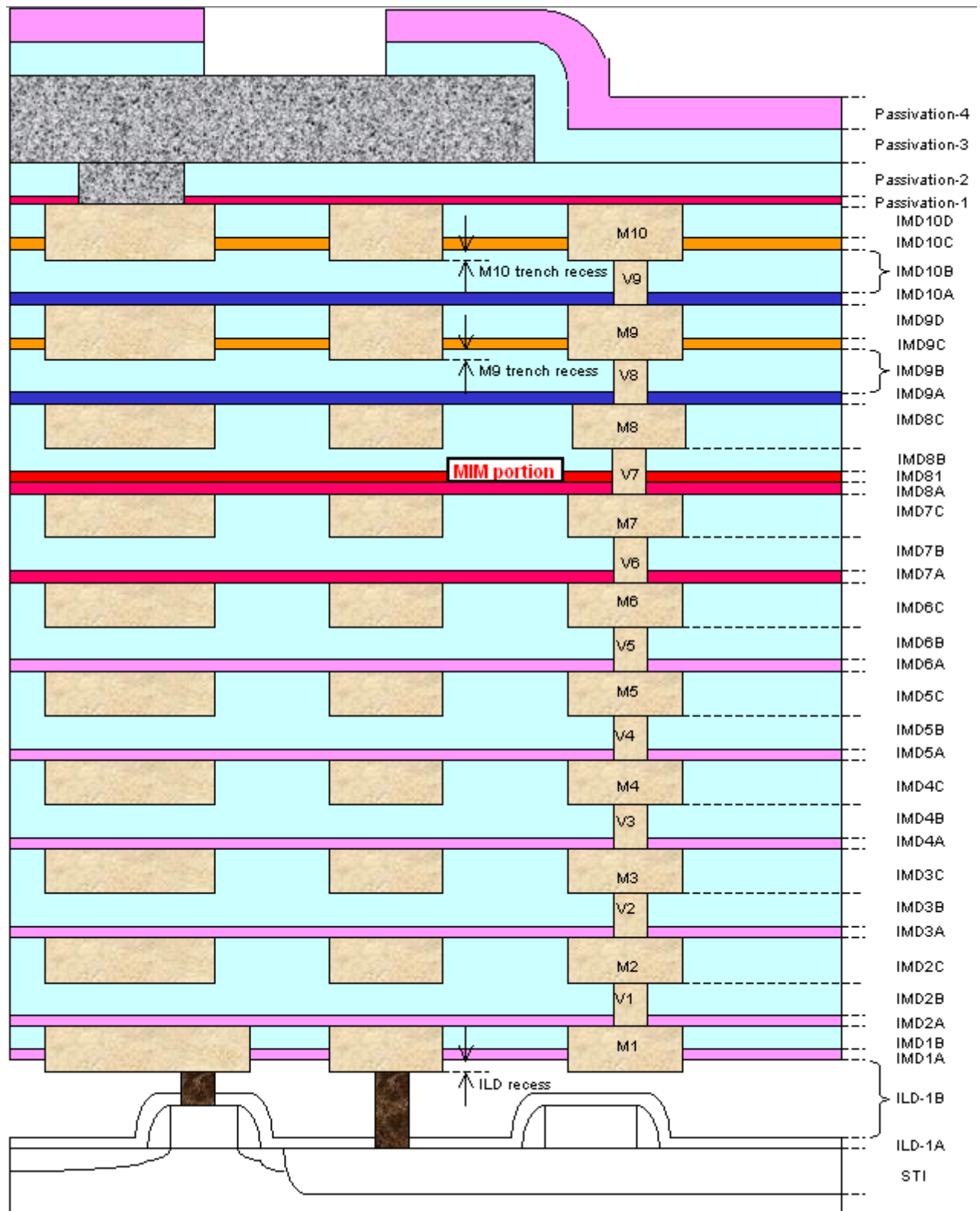
Part (I2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-411-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP

V0.1_P1

Part (I11):



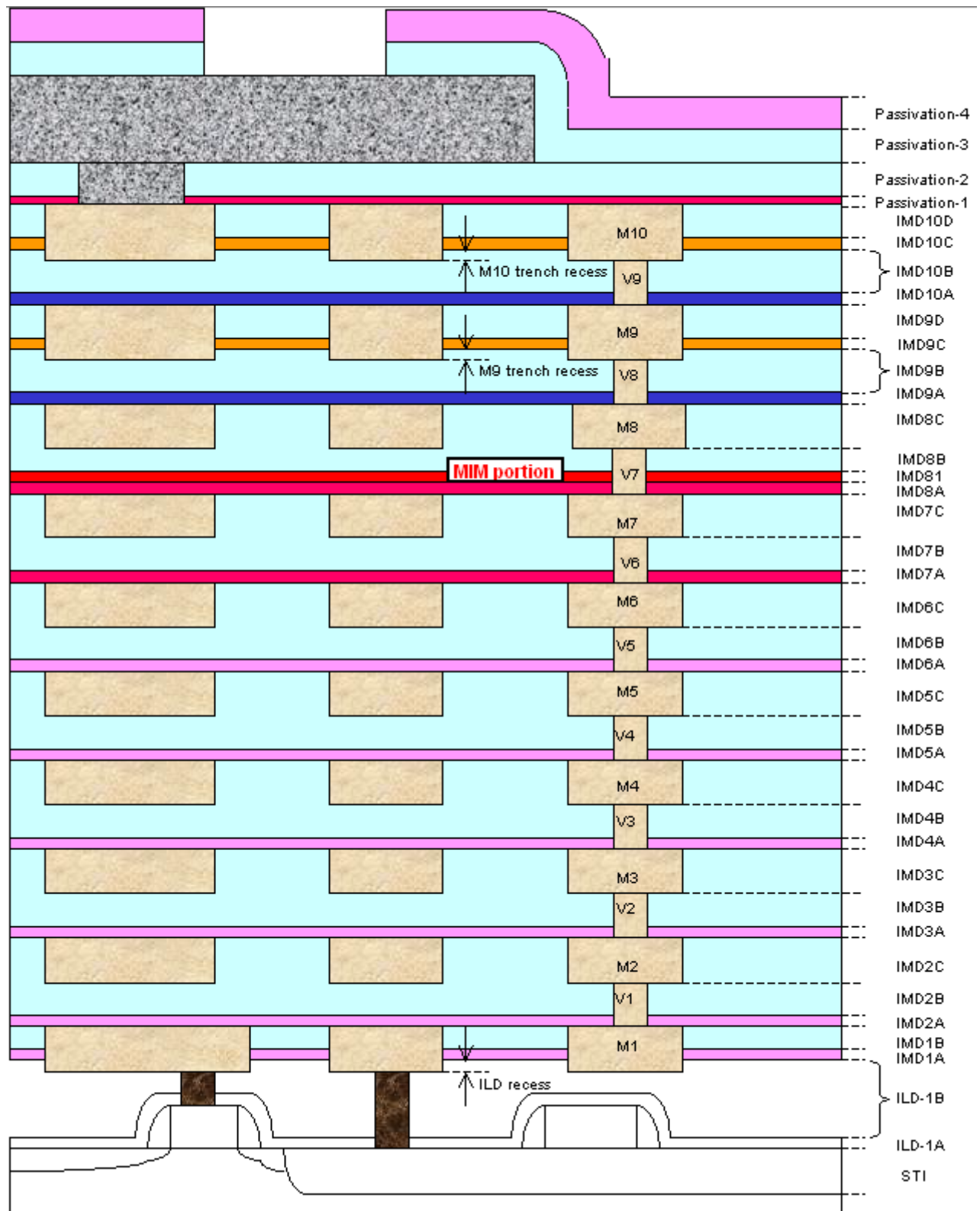
Part (I12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-412-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP

V0.1_P1

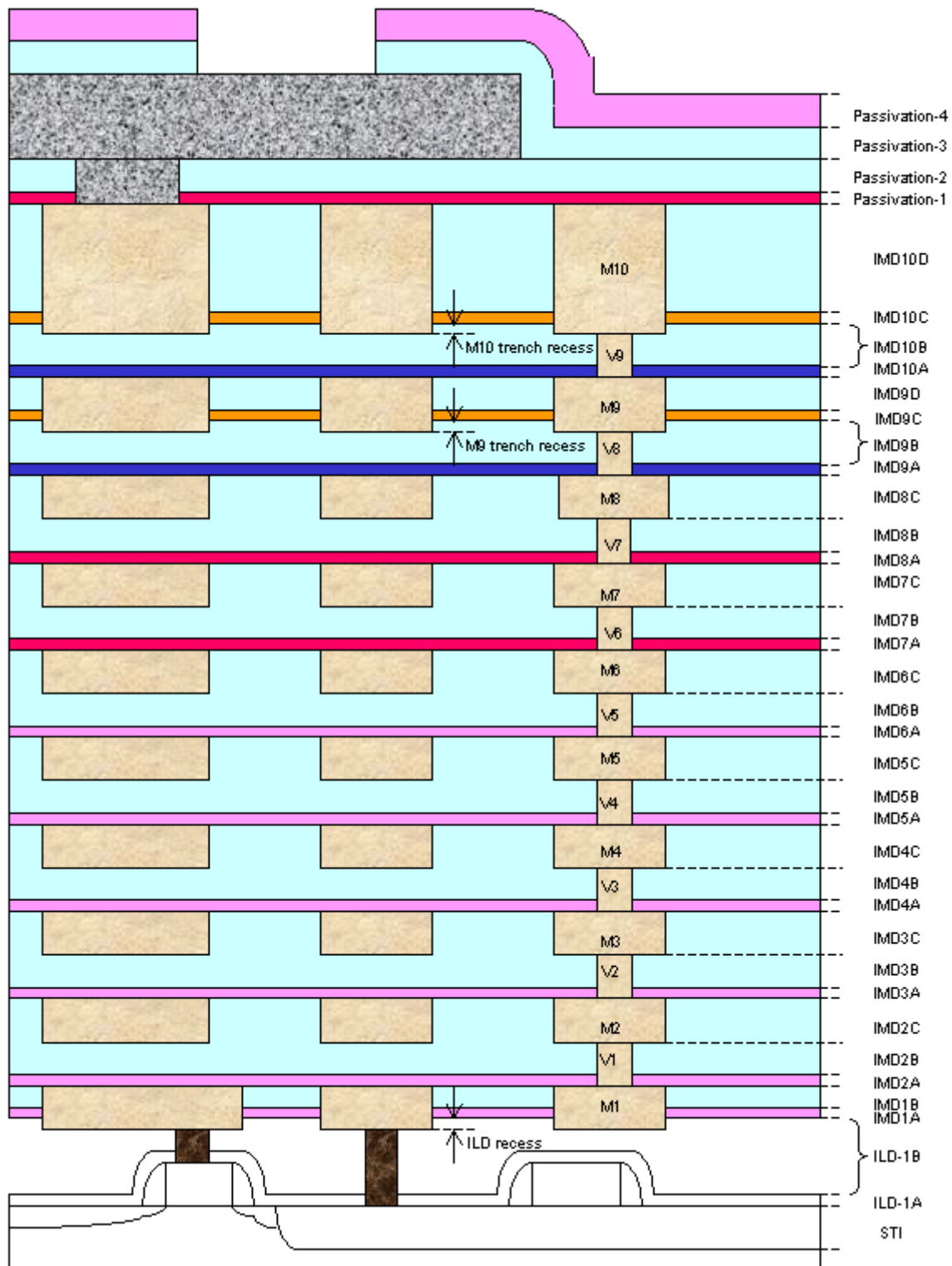
Part (I21):



Part (I22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

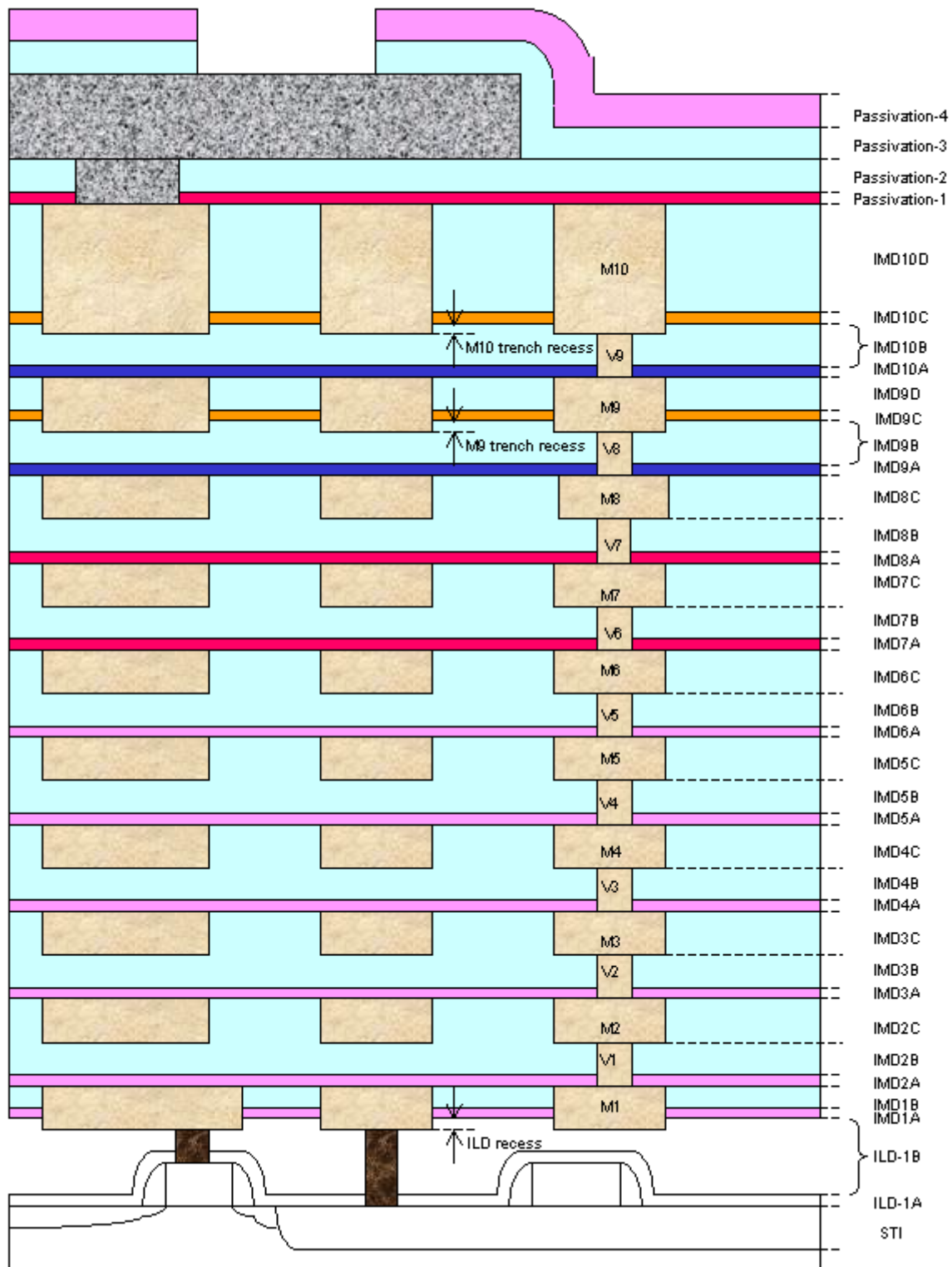
G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
Part (J1):



Part (J2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4J1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP
 V0.1_P1
 Part (J11):



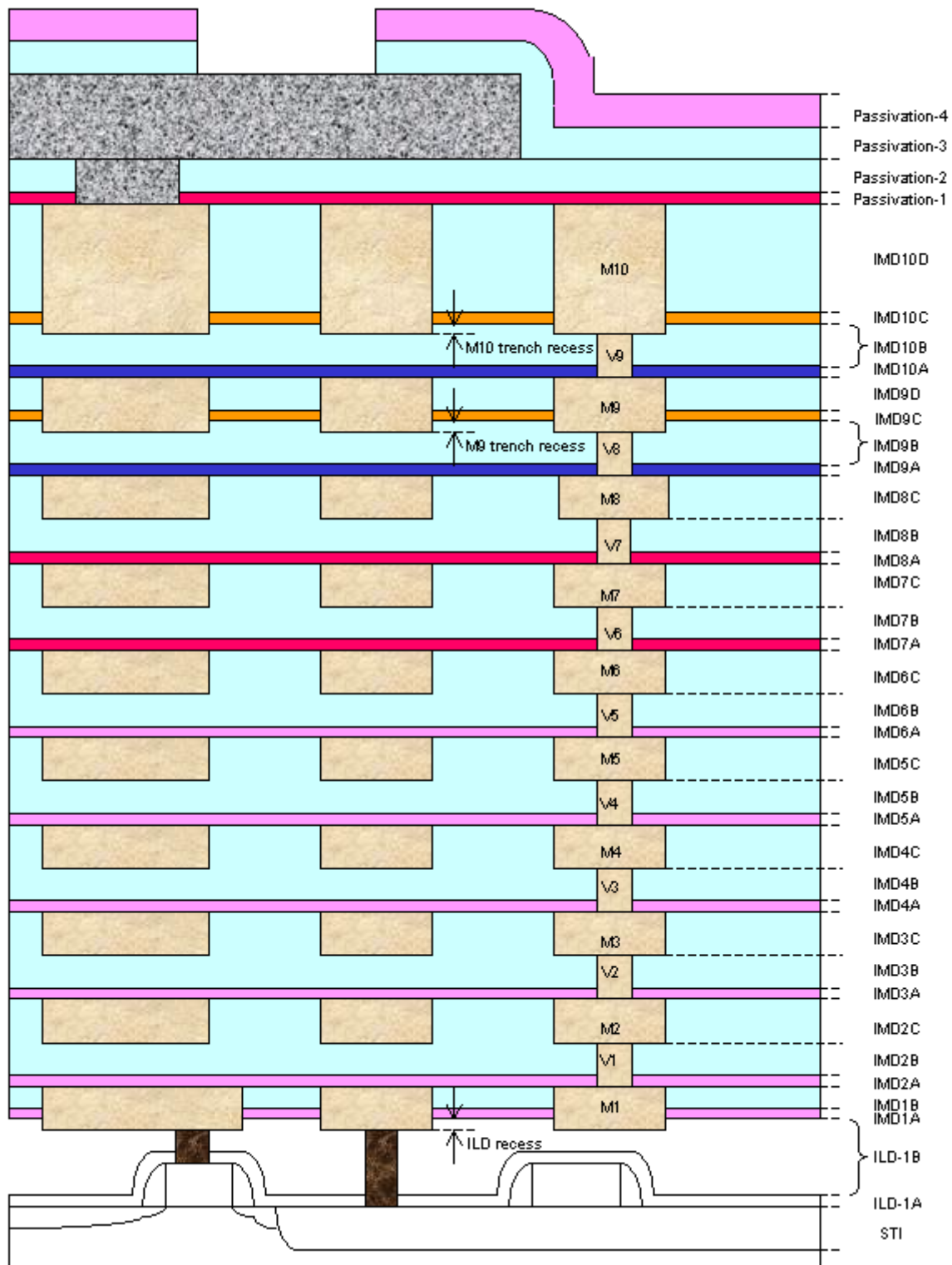
Part (J12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4J2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP

V0.1_P1

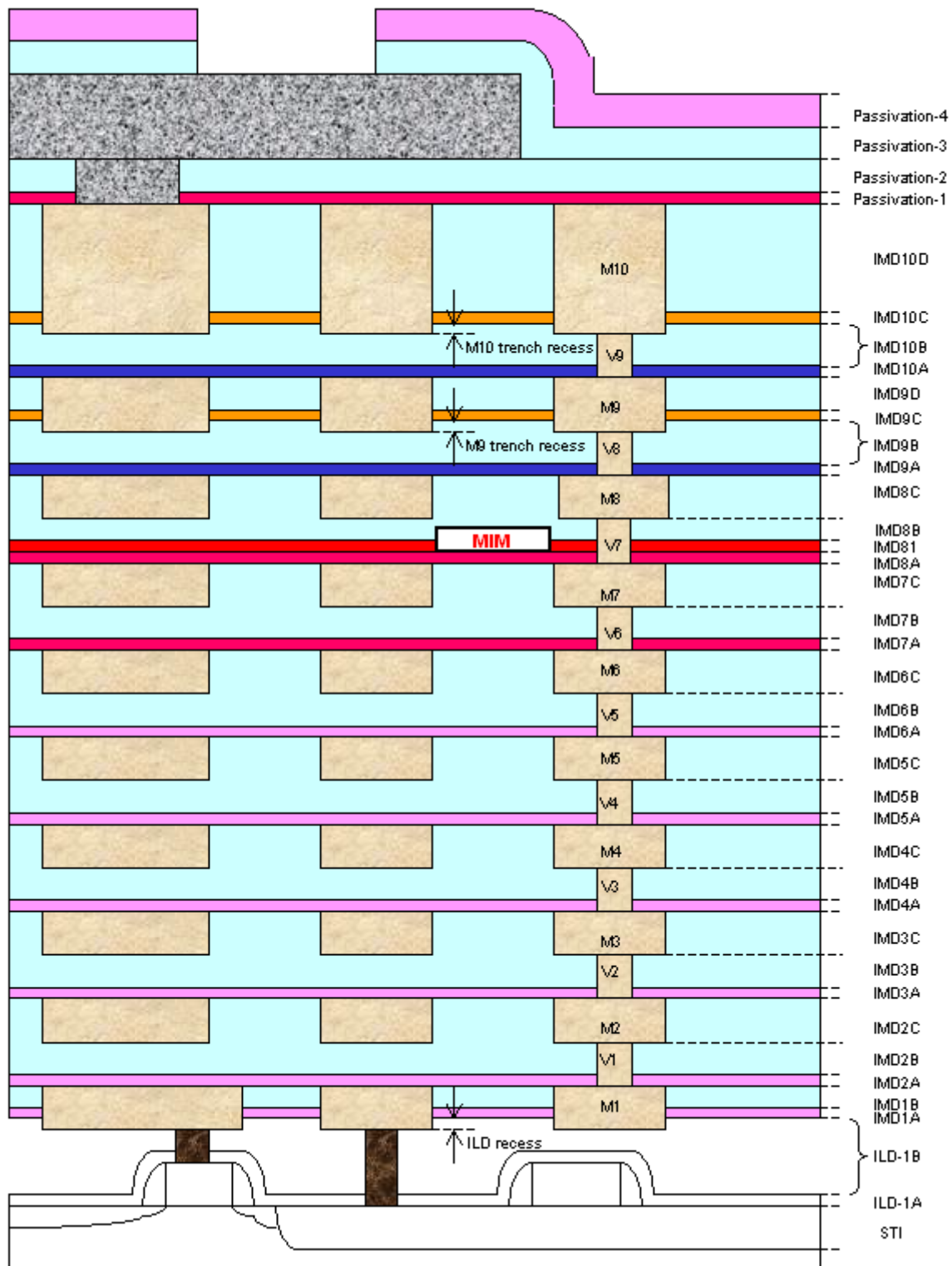
Part (J21):



Part (J22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

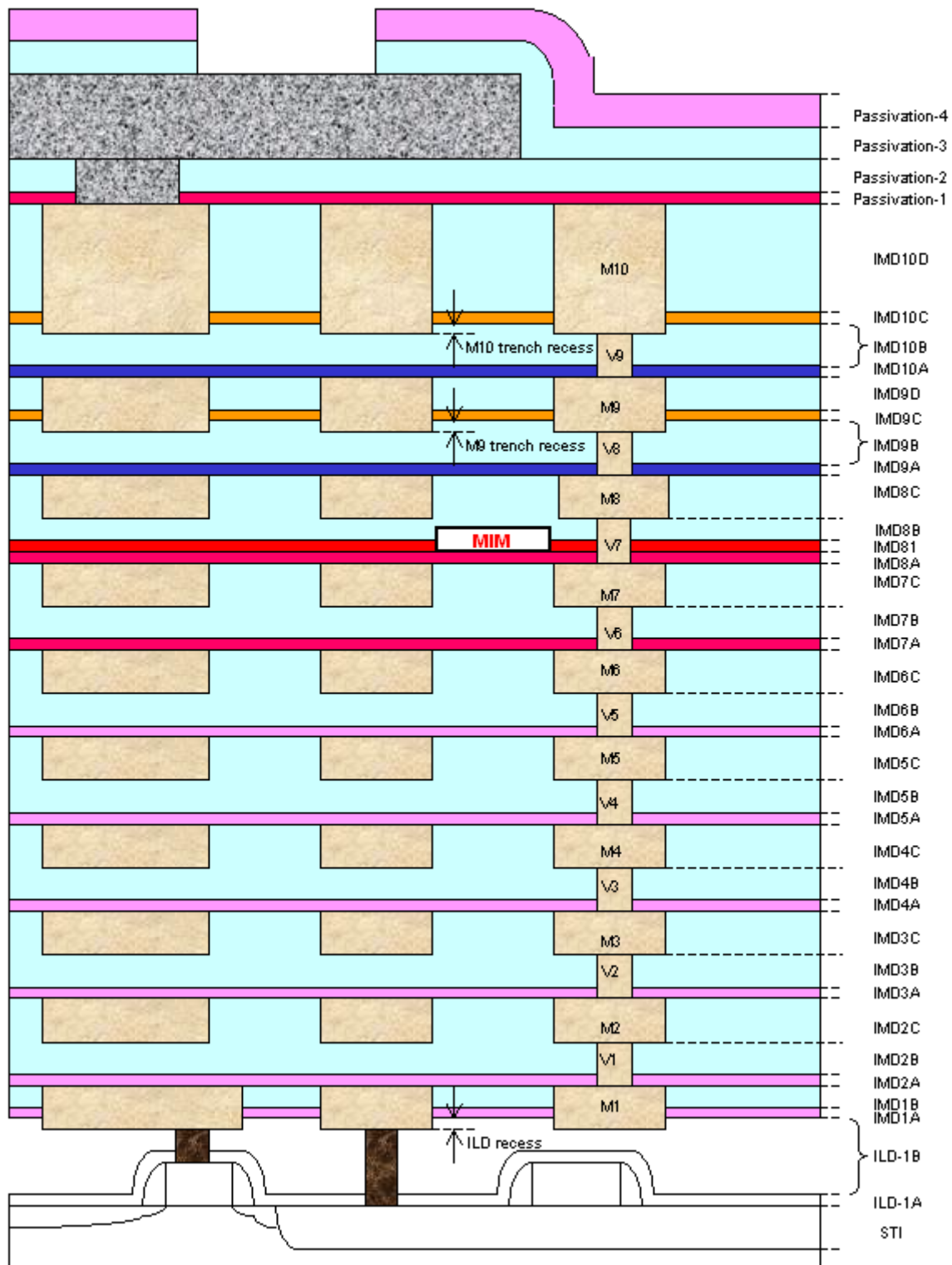
G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
Part (K1):



Part (K2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

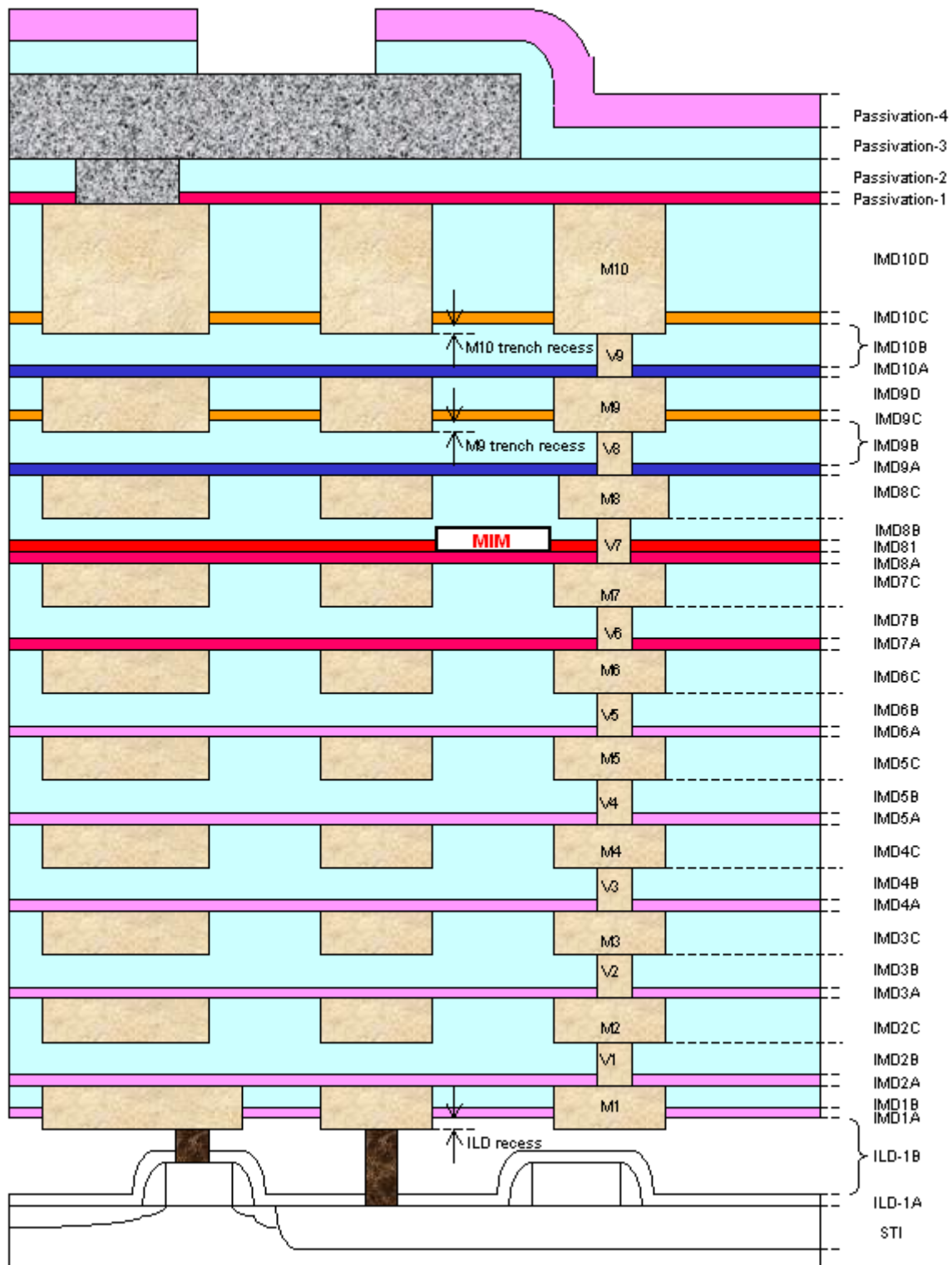
G-4K1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP
 V0.1_P1
 Part (K11):



Part (K12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

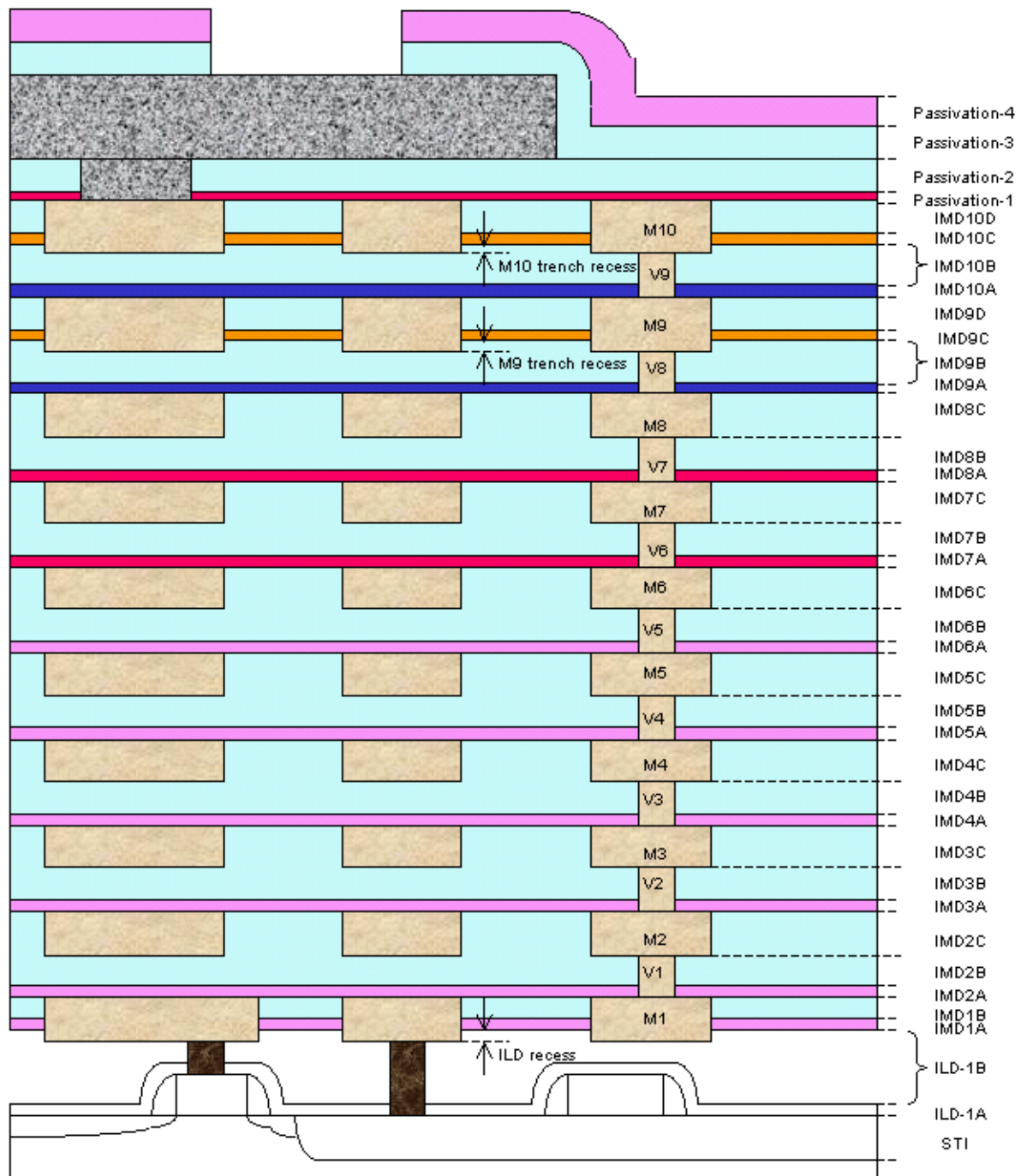
G-4K2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP
 V0.1_P1
 Part (K21):



Part (K22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

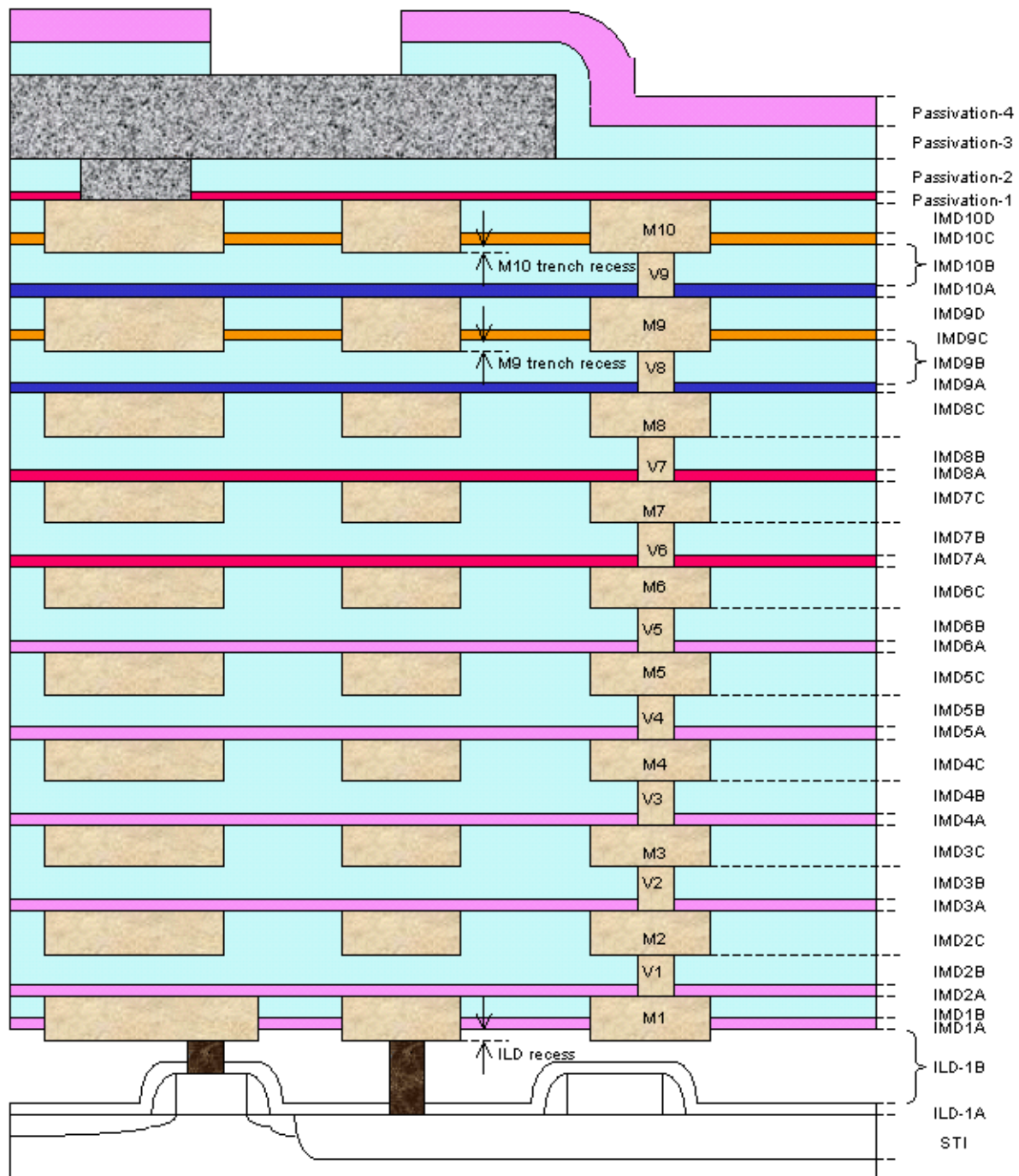
G-4L-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1
Part (L1):



Part (L2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
IMD10A	0.235	7
IMD10B	0.4	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

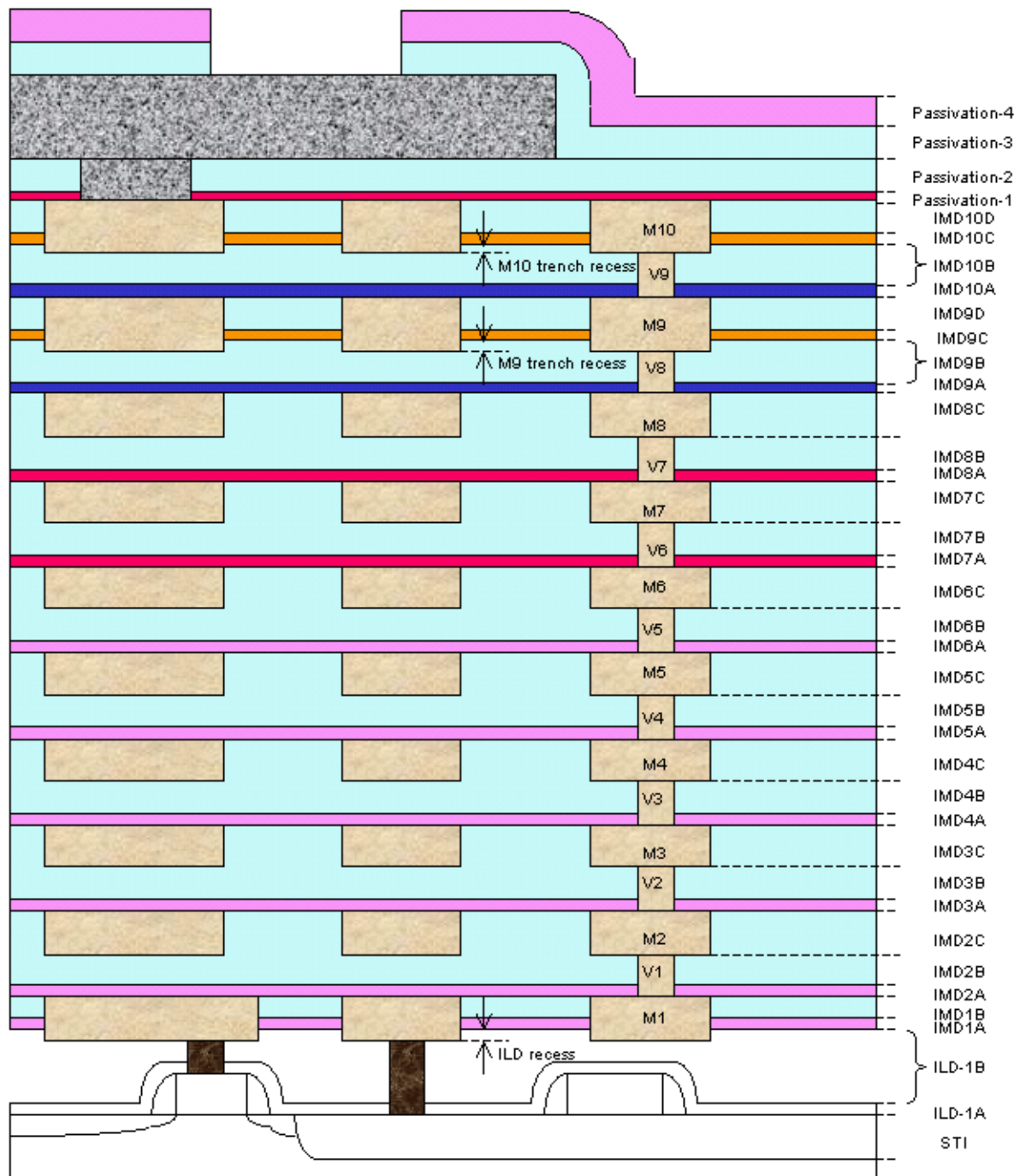
G-4L1-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP
V0.1_P1
Part (L11):



Part (L12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
IMD10A	0.235	7
IMD10B	0.4	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

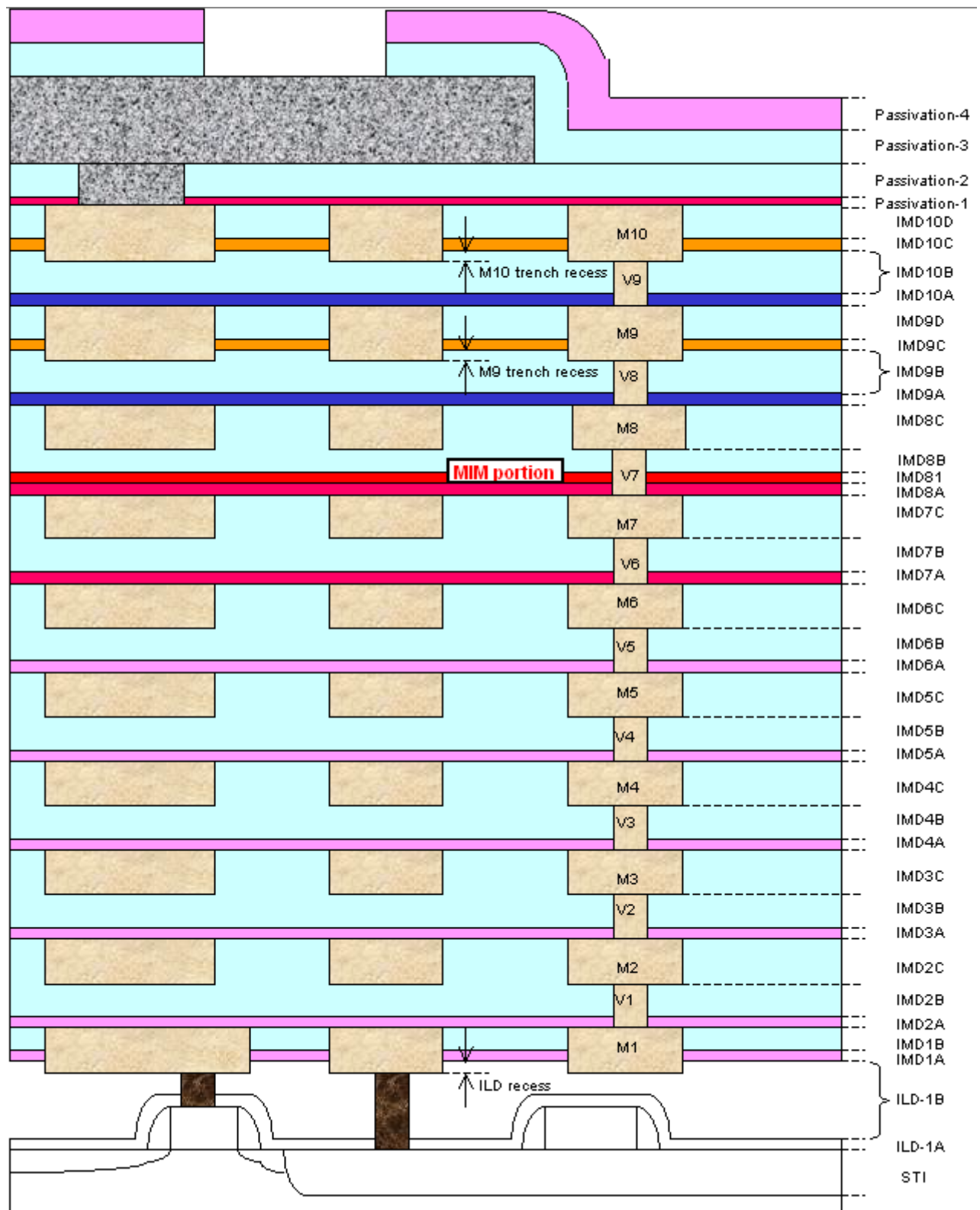
G-4L2-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP
 V0.1_P1
 Part (L21):



Part (L22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.235	7
IMD9B	0.4	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	1.14	3.7
IMD10A	0.235	7
IMD10B	0.4	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	1.14	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1
Part (M1):



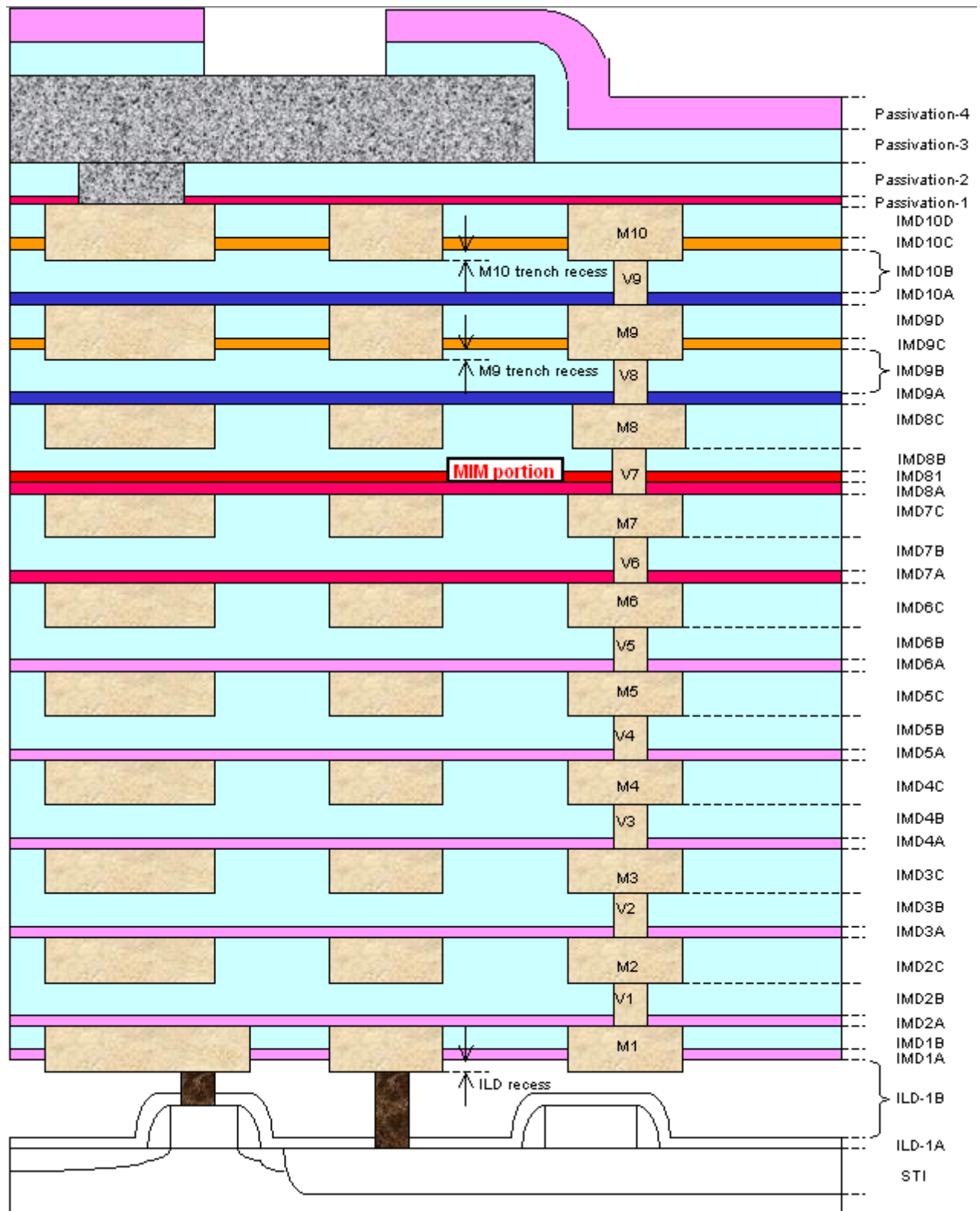
Part (M2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4M1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP

V0.1_P1

Part (M11):



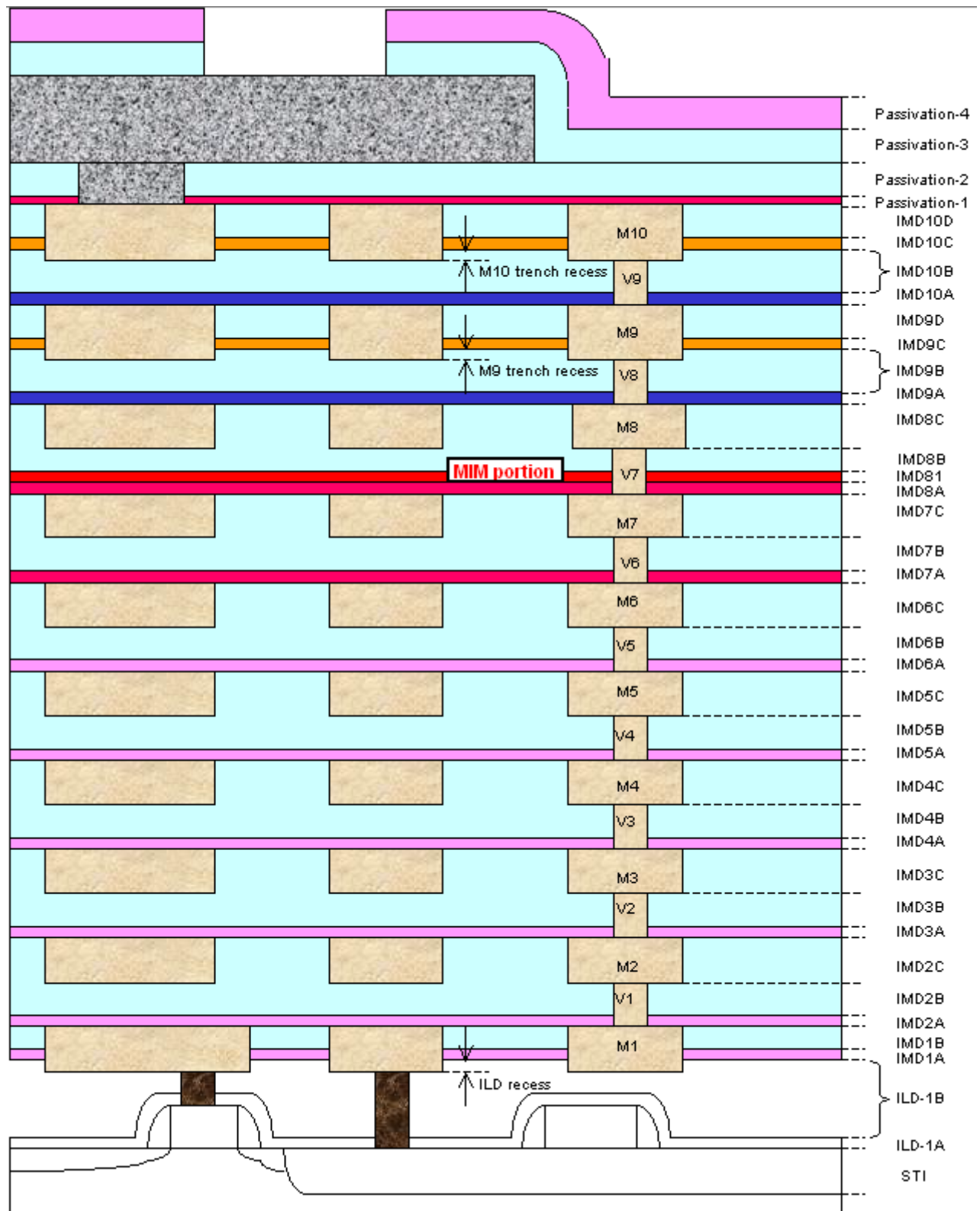
Part (M12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4M2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP

V0.1_P1

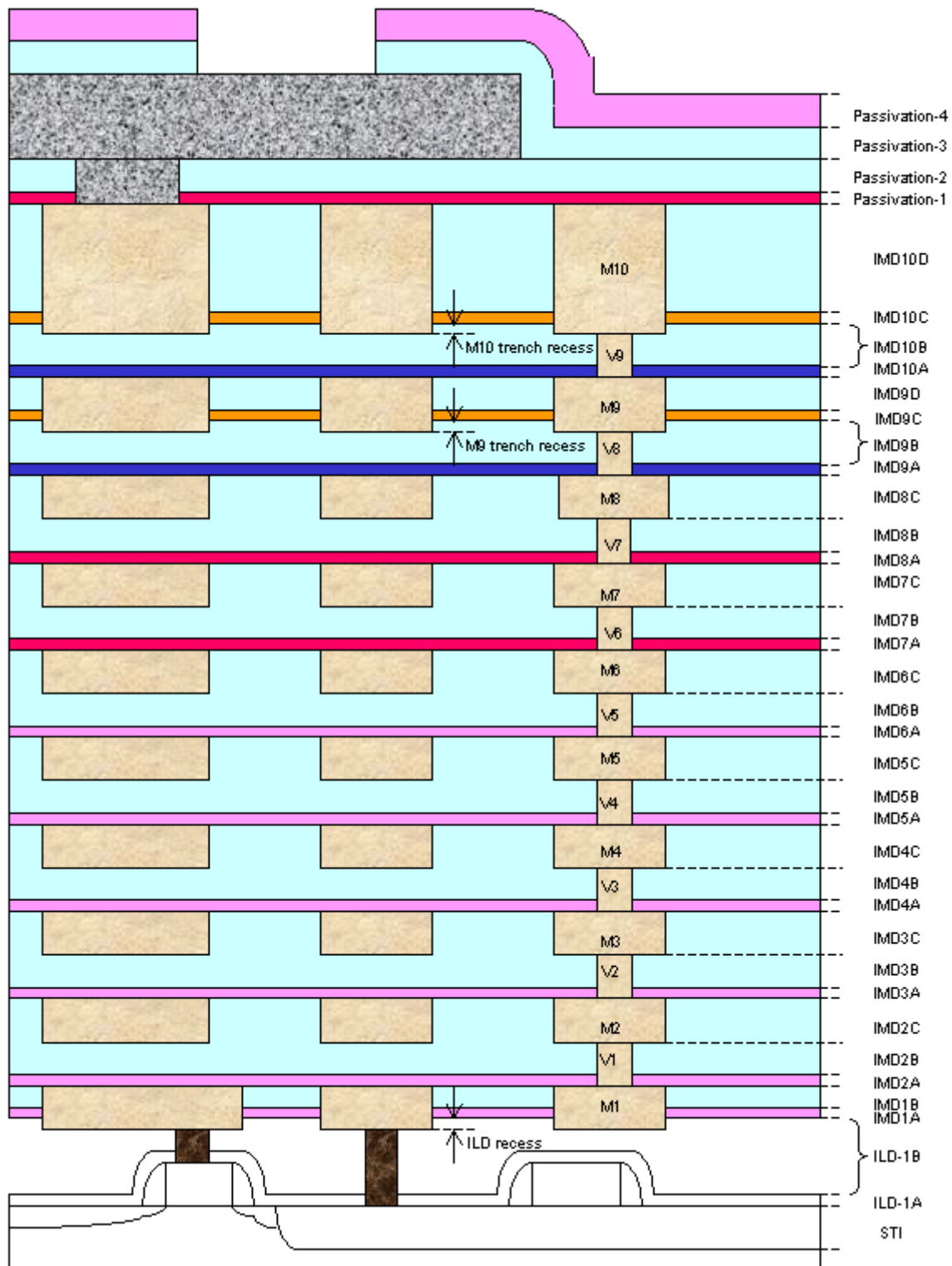
Part (M21):



Part (M22):

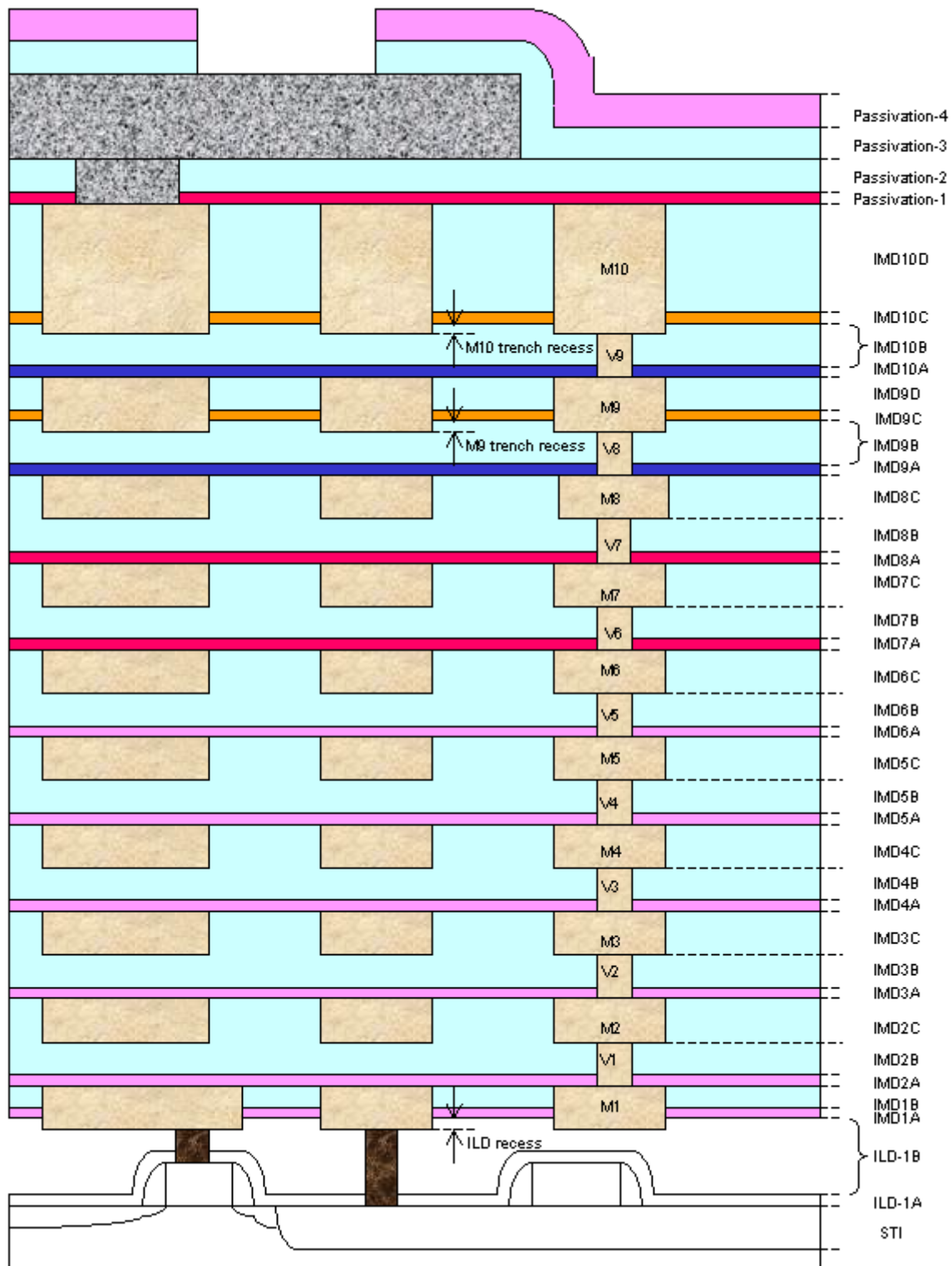
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.02	3.7
IMD10C	0.09	5
IMD10D	0.69	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
Part (N1):



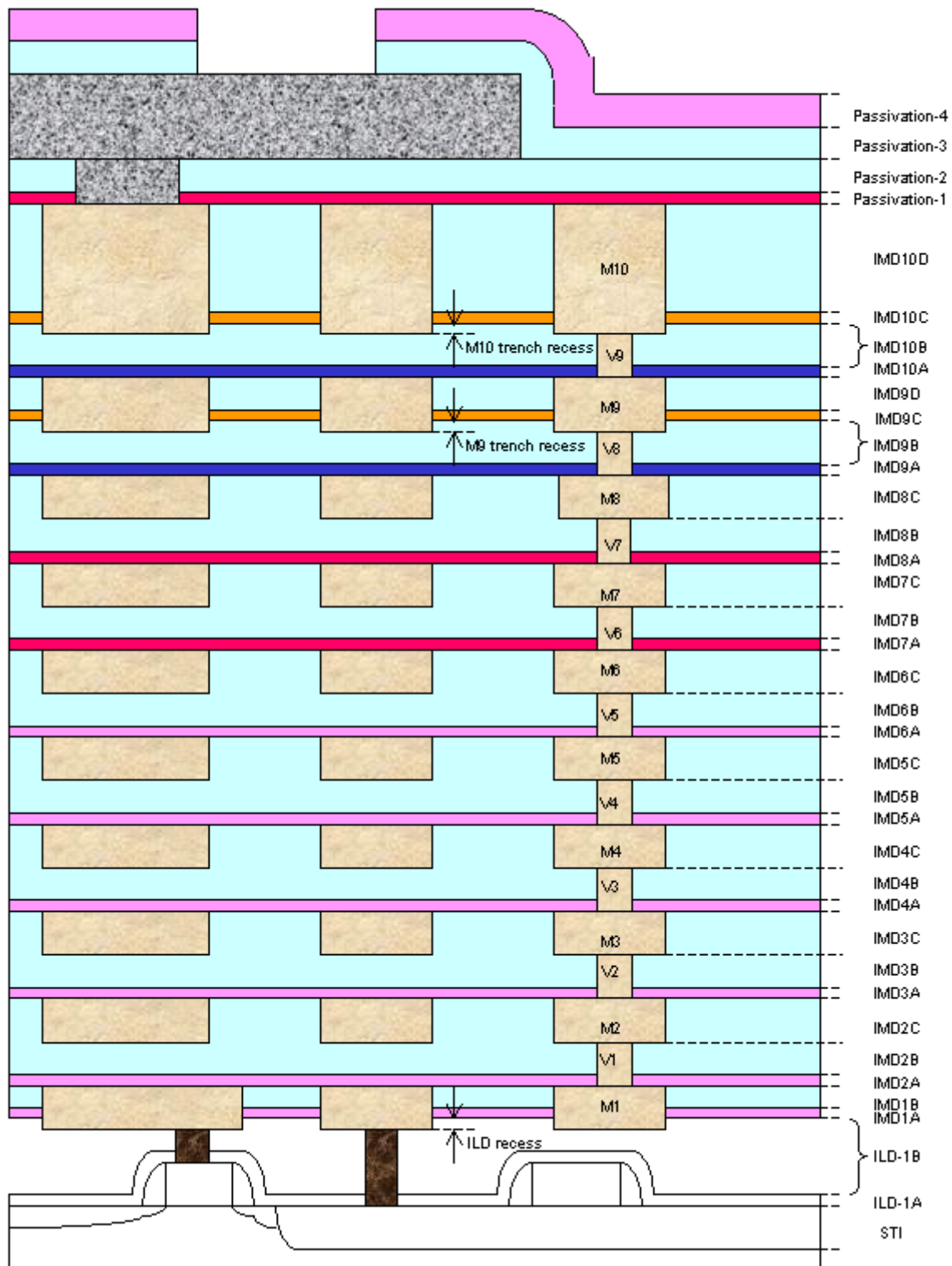
Part (N2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4N1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1
Part (N11):

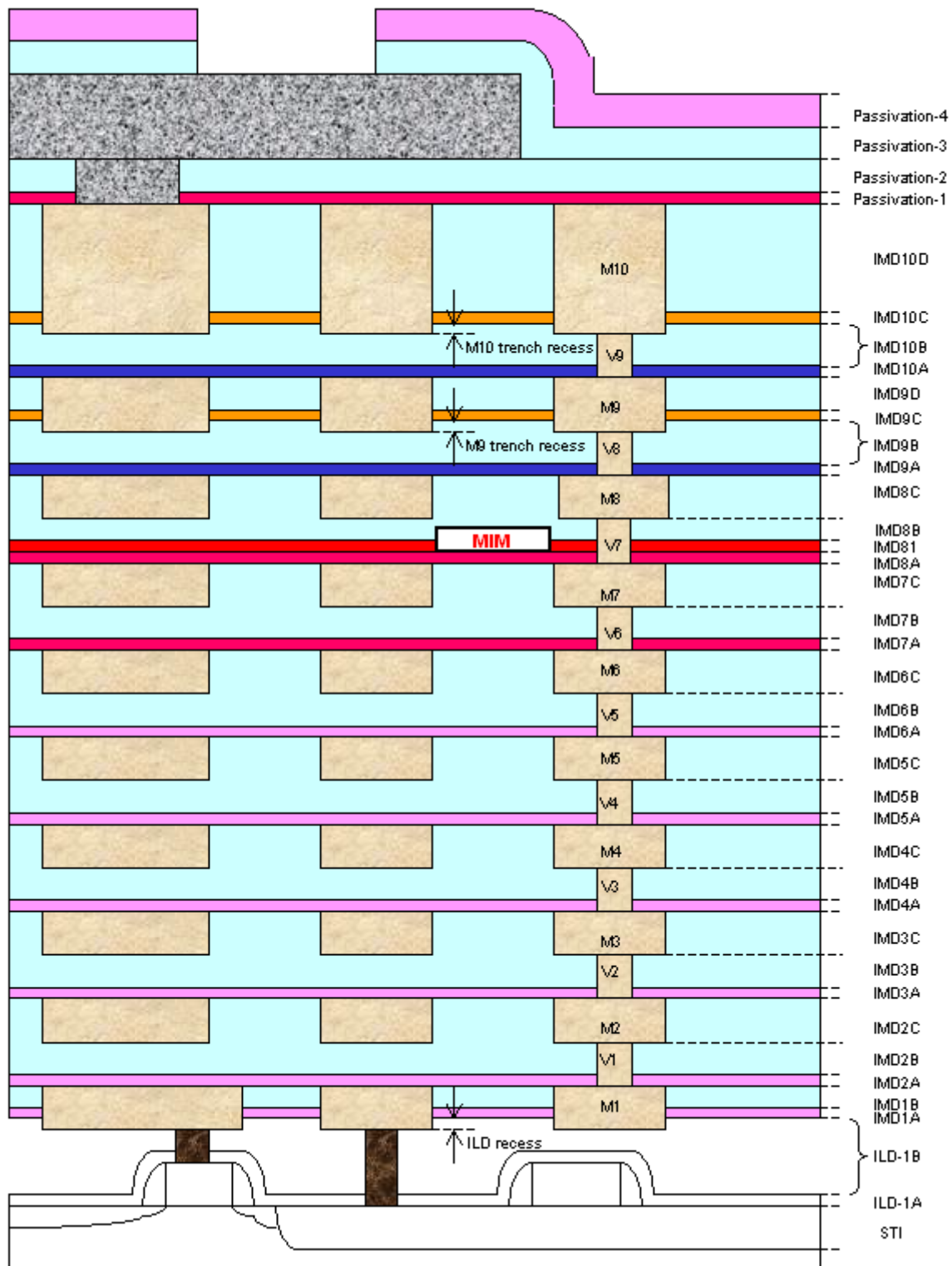
Part (N12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4N2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1
Part (N21):

Part (N22):

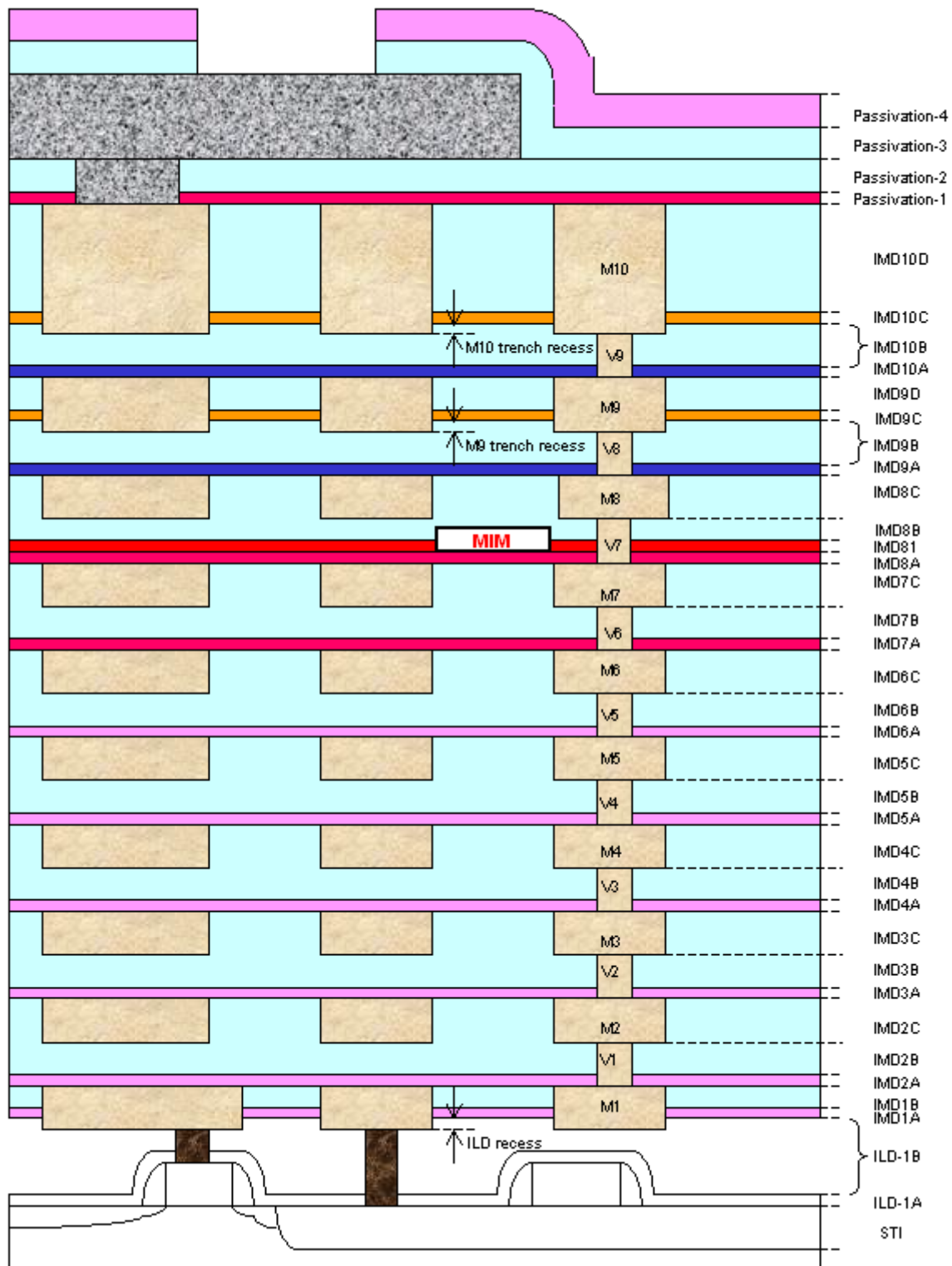
Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-40-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1
Part (O1):

Part (O2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

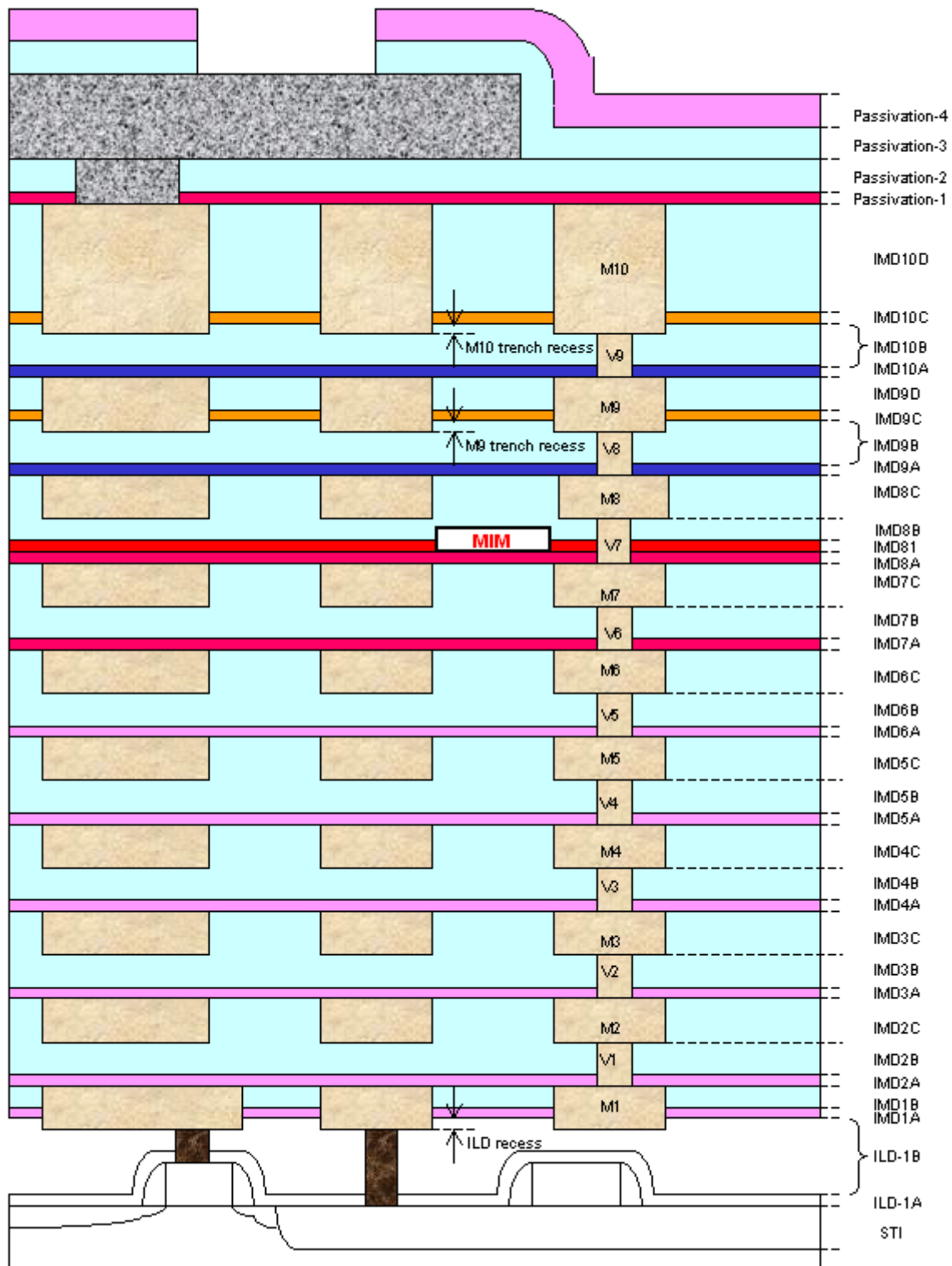
G-401-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1
Part (O11):



Part (O12):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-402-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1
Part (O21):

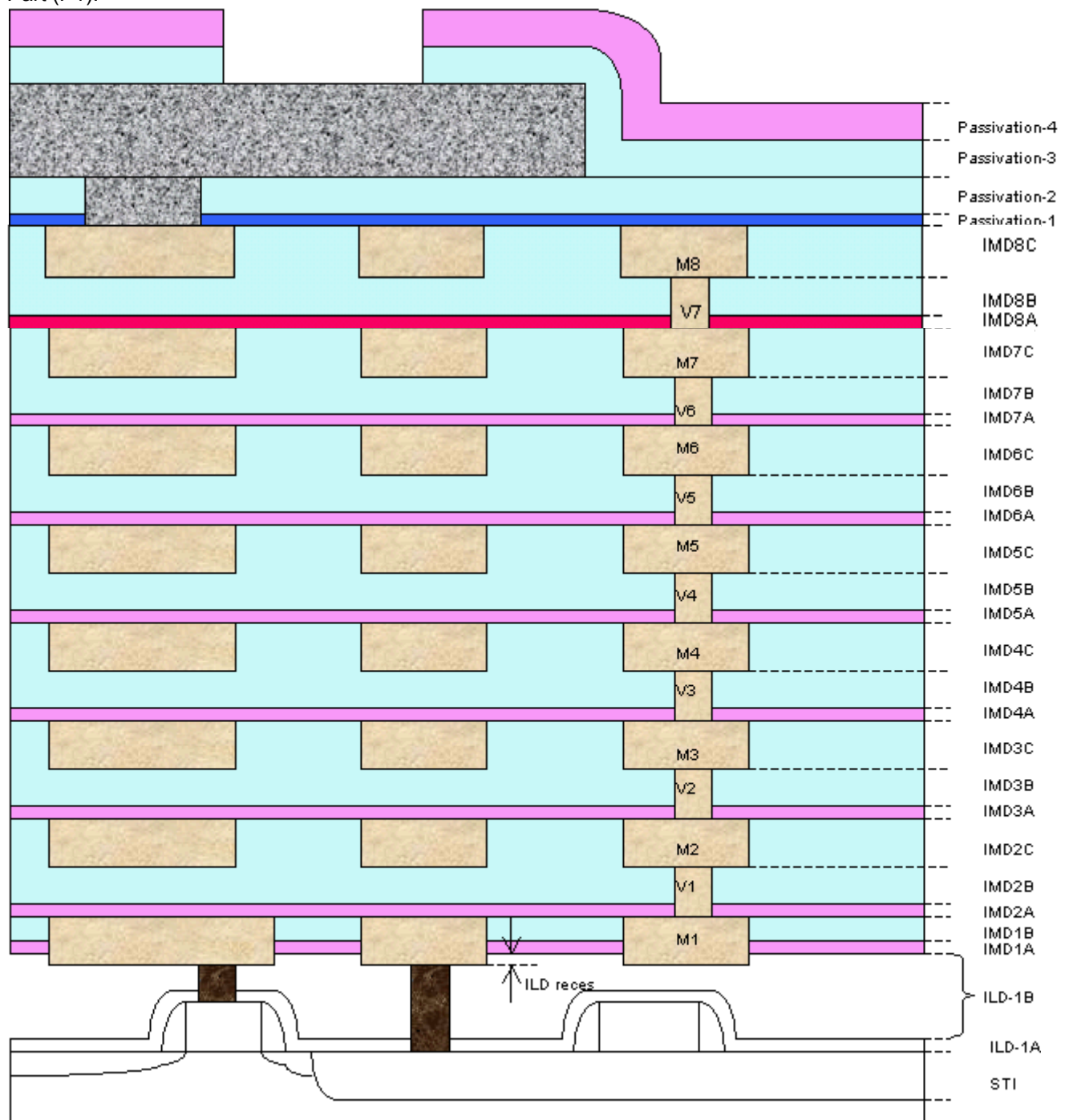


Part (O22):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.085	7
ILD1B	0.415	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.06	7
IMD7B	0.26	3.7
IMD7C	0.36	3.7
IMD8A	0.05	7
IMD81	0.035	7
IMD8B	0.51	3.7
IMD8C	0.36	3.7
IMD9A	0.135	7
IMD9B	0.5	3.7
M9 trench recess	0.02	3.7
IMD9C	0.09	5
IMD9D	0.69	3.7
IMD10A	0.135	7
IMD10B	0.5	3.7
M10 trench recess	0.07	3.7
IMD10C	0.2	5
IMD10D	2.98	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_METAL3.6K-INTERCAP V0.1_P1

Part (P1):



Part (P2):

Dielectric	Thickness (um)	Dielectric Constant
STI	0.35	4.1
ILD1A	0.038	7
ILD1B	0.462	4.1
ILD recess	0.03	4.1
IMD1A	0.03	5
IMD1B	0.12	2.9
IMD2A	0.05	5
IMD2B	0.11	2.9
IMD2C	0.22	2.9
IMD3A	0.05	5
IMD3B	0.11	2.9
IMD3C	0.22	2.9
IMD4A	0.05	5
IMD4B	0.11	2.9
IMD4C	0.22	2.9
IMD5A	0.05	5
IMD5B	0.11	2.9
IMD5C	0.22	2.9
IMD6A	0.05	5
IMD6B	0.11	2.9
IMD6C	0.22	2.9
IMD7A	0.05	5
IMD7B	0.11	2.9
IMD7C	0.22	2.9
IMD8A	0.06	7
IMD8B	0.26	3.7
IMD8C	0.36	3.7
Passivation1	0.07	7
Passivation2	1.2	4.1
Passivation3	0.4	4.1
Passivation4	0.5	7

5. Look Up Tables

All dimensions in look-up table are based on drawing size and the simulation has been considered the process variation.

G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

L65_G4A_Wd[X]X.typ : typical case.

L65_G4A_Wd[X]X.min : minimum case.

L65_G4A_Wd[X]X.max : maximum case.

L65_G4A_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4A_Wd[X]X.worst : RC maximum case.

L65_G4A_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4A1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

L65_G4A1_Wd[X]X.typ : typical case.

L65_G4A1_Wd[X]X.min : minimum case.

L65_G4A1_Wd[X]X.max : maximum case.

L65_G4A1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4A1_Wd[X]X.worst : RC maximum case.

L65_G4A1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4A2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

L65_G4A2_Wd[X]X.typ : typical case.

L65_G4A2_Wd[X]X.min : minimum case.

L65_G4A2_Wd[X]X.max : maximum case.

L65_G4A2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4A2_Wd[X]X.worst : RC maximum case.

L65_G4A2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1

L65_G4B_Wd[X]X.typ : typical case.

L65_G4B_Wd[X]X.min : minimum case.

L65_G4B_Wd[X]X.max : maximum case.

L65_G4B_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4B_Wd[X]X.worst : RC maximum case.

L65_G4B_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4B1-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1

L65_G4B1_Wd[X]X.typ : typical case.

L65_G4B1_Wd[X]X.min : minimum case.

L65_G4B1_Wd[X]X.max : maximum case.

L65_G4B1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4B1_Wd[X]X.worst : RC maximum case.

L65_G4B1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4B2-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1

L65_G4B2_Wd[X]X.typ : typical case.

L65_G4B2_Wd[X]X.min : minimum case.

L65_G4B2_Wd[X]X.max : maximum case.

L65_G4B2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4B2_Wd[X]X.worst : RC maximum case.

L65_G4B2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

L65_G4C_Wd[X]X.typ : typical case.

L65_G4C_Wd[X]X.min : minimum case.

L65_G4C_Wd[X]X.max : maximum case.

L65_G4C_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4C_Wd[X]X.worst : RC maximum case.

L65_G4C_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V1.0_P1

L65_G4D_Wd[X]X.typ : typical case.

L65_G4D_Wd[X]X.min : minimum case.

L65_G4D_Wd[X]X.max : maximum case.

L65_G4D_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4D_Wd[X]X.worst : RC maximum case.

L65_G4D_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

L65_G4E_Wd[X]X.typ : typical case.

L65_G4E_Wd[X]X.min : minimum case.

L65_G4E_Wd[X]X.max : maximum case.

L65_G4E_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4E_Wd[X]X.worst : RC maximum case.

L65_G4E_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4E1-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

L65_G4E1_Wd[X]X.typ : typical case.

L65_G4E1_Wd[X]X.min : minimum case.

L65_G4E1_Wd[X]X.max : maximum case.

L65_G4E1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4E1_Wd[X]X.worst : RC maximum case.

L65_G4E1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4E2-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

L65_G4E2_Wd[X]X.typ : typical case.

L65_G4E2_Wd[X]X.min : minimum case.

L65_G4E2_Wd[X]X.max : maximum case.

L65_G4E2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4E2_Wd[X]X.worst : RC maximum case.

L65_G4E2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

L65_G4F_Wd[X]X.typ : typical case.

L65_G4F_Wd[X]X.min : minimum case.

L65_G4F_Wd[X]X.max : maximum case.

L65_G4F_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4F_Wd[X]X.worst : RC maximum case.

L65_G4F_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

L65_G4G_Wd[X]X.typ : typical case.

L65_G4G_Wd[X]X.min : minimum case.

L65_G4G_Wd[X]X.max : maximum case.

L65_G4G_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4G_Wd[X]X.worst : RC maximum case.

L65_G4G_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4G1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

L65_G4G1_Wd[X]X.typ : typical case.

L65_G4G1_Wd[X]X.min : minimum case.

L65_G4G1_Wd[X]X.max : maximum case.

L65_G4G1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4G1_Wd[X]X.worst : RC maximum case.

L65_G4G1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4G2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

L65_G4G2_Wd[X]X.typ : typical case.

L65_G4G2_Wd[X]X.min : minimum case.

L65_G4G2_Wd[X]X.max : maximum case.

L65_G4G2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4G2_Wd[X]X.worst : RC maximum case.

L65_G4G2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1

L65_G4H_Wd[X]X.typ : typical case.

L65_G4H_Wd[X]X.min : minimum case.

L65_G4H_Wd[X]X.max : maximum case.

L65_G4H_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4H_Wd[X]X.worst : RC maximum case.

L65_G4H_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

L65_G4I_Wd[X]X.typ : typical case.

L65_G4I_Wd[X]X.min : minimum case.

L65_G4I_Wd[X]X.max : maximum case.

L65_G4I_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4I_Wd[X]X.worst : RC maximum case.

L65_G4I_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4I1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

L65_G4I1_Wd[X]X.typ : typical case.

L65_G4I1_Wd[X]X.min : minimum case.

L65_G4I1_Wd[X]X.max : maximum case.

L65_G4I1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4I1_Wd[X]X.worst : RC maximum case.

L65_G4I1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4I2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

L65_G4I2_Wd[X]X.typ : typical case.

L65_G4I2_Wd[X]X.min : minimum case.

L65_G4I2_Wd[X]X.max : maximum case.

L65_G4I2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4I2_Wd[X]X.worst : RC maximum case.

L65_G4I2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

L65_G4J_Wd[X]X.typ : typical case.

L65_G4J_Wd[X]X.min : minimum case.

L65_G4J_Wd[X]X.max : maximum case.

L65_G4J_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4J_Wd[X]X.worst : RC maximum case.

L65_G4J_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4J1-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

L65_G4J1_Wd[X]X.typ : typical case.

L65_G4J1_Wd[X]X.min : minimum case.

L65_G4J1_Wd[X]X.max : maximum case.

L65_G4J1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4J1_Wd[X]X.worst : RC maximum case.

L65_G4J1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4J2-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

L65_G4J2_Wd[X]X.typ : typical case.

L65_G4J2_Wd[X]X.min : minimum case.

L65_G4J2_Wd[X]X.max : maximum case.

L65_G4J2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4J2_Wd[X]X.worst : RC maximum case.

L65_G4J2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

L65_G4K_Wd[X]X.typ : typical case.

L65_G4K_Wd[X]X.min : minimum case.

L65_G4K_Wd[X]X.max : maximum case.

L65_G4K_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4K_Wd[X]X.worst : RC maximum case.

L65_G4K_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4K1-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

L65_G4K1_Wd[X]X.typ : typical case.

L65_G4K1_Wd[X]X.min : minimum case.

L65_G4K1_Wd[X]X.max : maximum case.

L65_G4K1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4K1_Wd[X]X.worst : RC maximum case.

L65_G4K1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4K2-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

L65_G4K2_Wd[X]X.typ : typical case.

L65_G4K2_Wd[X]X.min : minimum case.

L65_G4K2_Wd[X]X.max : maximum case.

L65_G4K2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4K2_Wd[X]X.worst : RC maximum case.

L65_G4K2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4L-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V1.1_P1

L65_G4L_Wd[X]X.typ : typical case.

L65_G4L_Wd[X]X.min : minimum case.

L65_G4L_Wd[X]X.max : maximum case.

L65_G4L_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4L_Wd[X]X.worst : RC maximum case.

L65_G4L_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4L1-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL25K-INTERCAP V0.1_P1

L65_G4L1_Wd[X]X.typ : typical case.

L65_G4L1_Wd[X]X.min : minimum case.

L65_G4L1_Wd[X]X.max : maximum case.

L65_G4L1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4L1_Wd[X]X.worst : RC maximum case.

L65_G4L1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4L2-LOGIC/MIXED_MODE65N-1P10M2T2H-LL/LOW_K/TOP_METAL12.5K/AL36K-INTERCAP V0.1_P1

L65_G4L2_Wd[X]X.typ : typical case.

L65_G4L2_Wd[X]X.min : minimum case.

L65_G4L2_Wd[X]X.max : maximum case.

L65_G4L2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4L2_Wd[X]X.worst : RC maximum case.

L65_G4L2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.1_P1

L65_G4M_Wd[X]X.typ : typical case.

L65_G4M_Wd[X]X.min : minimum case.

L65_G4M_Wd[X]X.max : maximum case.

L65_G4M_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4M_Wd[X]X.worst : RC maximum case.

L65_G4M_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4M1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL25K-INTERCAP V0.1_P1

L65_G4M1_Wd[X]X.typ : typical case.

L65_G4M1_Wd[X]X.min : minimum case.

L65_G4M1_Wd[X]X.max : maximum case.

L65_G4M1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4M1_Wd[X]X.worst : RC maximum case.

L65_G4M1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4M2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K/AL36K-INTERCAP V0.1_P1

L65_G4M2_Wd[X]X.typ : typical case.

L65_G4M2_Wd[X]X.min : minimum case.

L65_G4M2_Wd[X]X.max : maximum case.

L65_G4M2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4M2_Wd[X]X.worst : RC maximum case.

L65_G4M2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

L65_G4N_Wd[X]X.typ : typical case.

L65_G4N_Wd[X]X.min : minimum case.

L65_G4N_Wd[X]X.max : maximum case.

L65_G4N_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4N_Wd[X]X.worst : RC maximum case.

L65_G4N_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4N1-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

L65_G4N1_Wd[X]X.typ : typical case.

L65_G4N1_Wd[X]X.min : minimum case.

L65_G4N1_Wd[X]X.max : maximum case.

L65_G4N1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4N1_Wd[X]X.worst : RC maximum case.

L65_G4N1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4N2-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

L65_G4N2_Wd[X]X.typ : typical case.

L65_G4N2_Wd[X]X.min : minimum case.

L65_G4N2_Wd[X]X.max : maximum case.

L65_G4N2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4N2_Wd[X]X.worst : RC maximum case.

L65_G4N2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4O-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.1_P1

L65_G4O_Wd[X]X.typ : typical case.

L65_G4O_Wd[X]X.min : minimum case.

L65_G4O_Wd[X]X.max : maximum case.

L65_G4O_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4O_Wd[X]X.worst : RC maximum case.

L65_G4O_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4O1-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL25K-INTERCAP V0.1_P1

L65_G4O1_Wd[X]X.typ : typical case.

L65_G4O1_Wd[X]X.min : minimum case.

L65_G4O1_Wd[X]X.max : maximum case.

L65_G4O1_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4O1_Wd[X]X.worst : RC maximum case.

L65_G4O1_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4O2-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K/AL36K-INTERCAP V0.1_P1

L65_G4O2_Wd[X]X.typ : typical case.

L65_G4O2_Wd[X]X.min : minimum case.

L65_G4O2_Wd[X]X.max : maximum case.

L65_G4O2_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4O2_Wd[X]X.worst : RC maximum case.

L65_G4O2_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_METAL3.6K-INTERCAP V0.1_P1

L65_G4P_Wd[X]X.typ : typical case.

L65_G4P_Wd[X]X.min : minimum case.

L65_G4P_Wd[X]X.max : maximum case.

L65_G4P_Wd[X]X.cmp : comparison between typical, minimum and maximum cases.

L65_G4P_Wd[X]X.worst : RC maximum case.

L65_G4P_Wd[X]X.best : RC minimum case.

Note: [X] means 1, 2, 3 or 4.

6. Appendix Revision History

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
T.0	1	08/25/2005	-	-	Original
T.1	1	11/25/2005	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP VT.0_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL12.5K-INTERCAP VT.0_P1 3. No 4.3 and 4.4 section for WEE and Rs tables. 4. ILD1B thickness = 0.315um for G4A and G4B. 5. AL-Pad thickness = 0.8um.	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP VT.1_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL12.5K-INTERCAP VT.1_P1 3. Add 4.3 and 4.4 section for WEE and Rs tables into this document. 4. ILD1B thickness = 0.415um for G4A and G4B. 5. AL-Pad thickness = 1.2um.	Add WEE and Rs tables into this model.
0.1	1	05/12/2006	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP VT.1_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL12.5K-INTERCAP VT.1_P1 3. 1X-Metal WEE and Rs table of G4A and G4B. 4. Lookup tables of G4A and G4B.	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP VT.2_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K-INTERCAP V0.1_P1 3. Update 1X-Metal WEE and Rs table of G4A and G4B. 4. Update lookup tables of G4A and G4B.	Based on three wafers C and R mapping data of KW856(#8) and KCS24(#8 and #10), the version of G4B is upgraded to 0.1 from T.1.
0.2	1	06/20/2006	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP VT.2_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K-INTERCAP V0.1_P1 3. No G-4C~G-4F. 4. There are not WEE and Rs tables for 2X-metal and 4X-metal. 5. Lookup tables.	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP V0.1_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K-INTERCAP V0.2_P1 3. G-4C-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.1_P1 G-4D-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V0.1_P1 G-4E-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL8K-INTERCAP V0.1_P1 G-4F-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.1_P1 4. Add WEE and Rs tables for 2X-metal and 4X-metal. 5. Update lookup tables.	1. Based on two wafers C and R mapping data of KSS31 (#19 and #20). Overall subset models are upgraded to 0.X version. 2. Add WEE and Rs tables for 2X-metal and 4X-metal.

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
0.3	1	11/14/2006	1. Change the naming. G-4A-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K-INTERCAP V0.1_P1 G-4B-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K-INTERCAP V0.2_P1 G-4C-LOGIC/MIXED_MODE65N-1P10M-LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.1_P1 G-4D-LOGIC/MIXED_MODE65N-1P9M2H-LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V0.1_P1 G-4E-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL8K-INTERCAP V0.1_P1 G-4F-LOGIC/MIXED_MODE65N-1P9M2F-LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.1_P1 2. Poly bias of G-4A and G-4C: 0.005um. 3. ILD1A thickness of G-4A and G-4C: 0.085um. 4. Metal and dielectric layer thickness variation: +/-10%. 5. No G-4G and G-4H. 6. Lookup tables.	1. Change the naming. G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V0.2_P1 G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V0.3_P1 G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P1 G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V0.2_P1 G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V0.2_P1 G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P1 2. Poly bias of G-4A and G-4C: 0.003um. 3. ILD1A thickness of G-4A and G-4C: 0.038um. 4. Metal and dielectric layer thickness variation: +/-15%. 5. Add G-4G and G-4H for L65 SP 1P10M intercap model. 6. Update whole lookup tables.	1. Update L65 LL intercap model for poly bias and ILD1A thickness changes. 2. Add two subset models (G-4G and G-4H) for L65 SP 1P10M intercap model.
0.3	2	12/18/2006	1. The table in section 3.1.	1. Update the table in section 3.1.	1. Modify typo error in the table of section 3.1.
0.3	3	02/09/2007	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V0.2_P1 2. G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P1 3. In G-4A and G-4C subset models, the thickness of ILD1B is 0.415um.	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V0.2_P2 2. G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P2 3. In G-4A and G-4C subset models, the thickness of ILD1B is 0.462um.	1. Modify typo error for ILD1B thickness in G-4A and G-4C.
0.4	1	03/07/2007	1. No G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V0.1P1 G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1 G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1	1. Add G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V0.1P1 G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1 G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1	Add G-4I~G-4K subset models for mixed-mode/RF application.

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0.5	1	04/24/2007	1. No G-4L-LOGIC/MIXED_MODE65N-1P10M-2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V0.1P1 2. G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V0.1P1 3. G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1 4. G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V0.1P1	1. Add G-4L-LOGIC/MIXED_MODE65N-1P10M-2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V0.1P1 2. G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V0.2P1 3. G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V0.2P1 4. G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V0.2P1	1. Add G-4L subset model for L65 LL 1P10M intercap model. 2. Modify IMD7A from 0.07 to 0.06 in G-4I. 3. Modify IMD7A and IMD8A from 0.07 to 0.06 in G-4J. 4. Modify IMD7A from 0.07 to 0.06 in G-4K.
1.0	1	03/04/2008	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V0.2_P2 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V0.3_P1 3. G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P2 4. G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V0.2_P1 5. G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V0.2_P1 6. G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.2_P1 7. G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V0.1_P1 8. G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V0.1_P1	1. G-4A-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 2. G-4B-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K-INTERCAP V1.0_P1 3. G-4C-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1 4. G-4D-LOGIC/MIXED_MODE65N-1P9M2H-SP/LOW_K/TOP_METAL12.5K/CU_PAD-INTERCAP V1.0_P1 5. G-4E-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 6. G-4F-LOGIC/MIXED_MODE65N-1P9M2F-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1 7. G-4G-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 8. G-4H-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL8K/CU_PAD-INTERCAP V1.0_P1	1.The items 1~12 only metal1 and 1X metal silicon data are provided for v1.0 model verification. 2.Add three subset models (G-4M and G-4N and G-4O) for L65 SP 1P10M intercap model.

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1.0	1	03/04/2008	9. G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V0.2P1 10. G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V0.2P1 11. G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V0.2P1 12. G-4L-LOGIC/MIXED_MODE65N-1P10M-2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V0.1P1 13. G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 14. G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 15. G-4O-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 16. 4.3 WEE Table (2) Metal2 ~ Metal6 of G4A, G4C, G4G, G4H, G4I, G4J, G4K and G4L: (3) Metal7 and Metal8 of G4A, G4C, G4G, G4H, G4I, G4J, G4K and G4L: (4) Metal9 and Metal10 of G4A, G4C, G4G, G4H and G4I : Metal8 and Metal9 of G4E and G4F: Metal9 of G4J and G4K: 4.4 Rs Table (2) Metal2 ~ Metal6 of G4A, G4C, G4G, G4H, G4I, G4J, G4K and G4L: (3) Metal7 and Metal8 of G4A, G4C, G4G, G4H, G4I, G4J, G4K and G4L: (4) Metal9 and Metal10 of G4A, G4C, G4G, G4H and G4I : Metal8 and Metal9 of G4E and G4F: Metal9 of G4J and G4K:	9. G-4I-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 10. G-4J-LOGIC/MIXED_MODE65N-1P10M-LL/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 11. G-4K-LOGIC/MIXED_MODE65N-1P10M-LL/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 12. G-4L-LOGIC/MIXED_MODE65N-1P10M-2T2H-LL/LOW_K/TOP_METAL12.5K-INTERCAP V1.0_P1 13. G-4M-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL8K-INTERCAP V1.0_P1 14. G-4N-LOGIC/MIXED_MODE65N-1P10M-SP/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 15. G-4O-LOGIC/MIXED_MODE65N-1P10M-SP/MIM/LOW_K/TOP_METAL32.5K-INTERCAP V1.0_P1 16. 4.3 WEE Table (2) Metal2 ~ Metal6 of G4A, G4C, G4G, G4H, G4I, G4J, G4K, G4L, G4M, G4N and G4O: (3) Metal7 and Metal8 of G4A, G4C, G4G, G4H, G4I, G4J, G4K, G4L, G4M, G4N and G4O: (4) Metal9 and Metal10 of G4A, G4C, G4G, G4H, G4I and G4M: Metal8 and Metal9 of G4E and G4F: Metal9 of G4J, G4K, G4N and G4O: 4.4 Rs Table (2) Metal2 ~ Metal6 of G4A, G4C, G4G, G4H, G4I, G4J, G4K, G4L, G4M, G4N and G4O: (3) Metal7 and Metal8 of G4A, G4C, G4G, G4H, G4I, G4J, G4K, G4L, G4M, G4N and G4O: (4) Metal9 and Metal10 of G4A, G4C, G4G, G4H, G4I and G4M: Metal8 and Metal9 of G4E and G4F: Metal9 of G4J, G4K, G4N and G4O:	

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
1.0	2	04/03/2009	Replace the according DSM naming at the table list . G-03-EDRAM65N-TLR	For SP G-03-EDRAM65N-SP/LOW_K-TLR For LL G-03-EDRAM65N-LL/LOW_K-TLR	
1.1	1	12/10/2010	For AL 12K subsets : minimum width=3um none none	For AL 12K subsets : minimum width=2.7um Add AL 25K subsets Add AL 36K subsets	
1.2	1	01/06/2011	No G-4P subset	Add G-4P subset : G-4P-LOGIC/MIXED_MODE65N-1P8M1T-LL/LOW_K/TOP_MET AL3.6K-INTERCAP V0.1_P1 Updated look up tables' name	

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